BIOLOGICAL MEMBRANES

Biological membranes are
the boundaries of cells or cellular organelles, they define the inside and the outside of a cell or an organelle

Biological membranes are composed by: a bilayer of Polar Lipids, that act as barrier to the permeability and Proteins, that act as a transport system of pumps and channels. Membranes also contain carbohydrates linked to lipids and proteins.

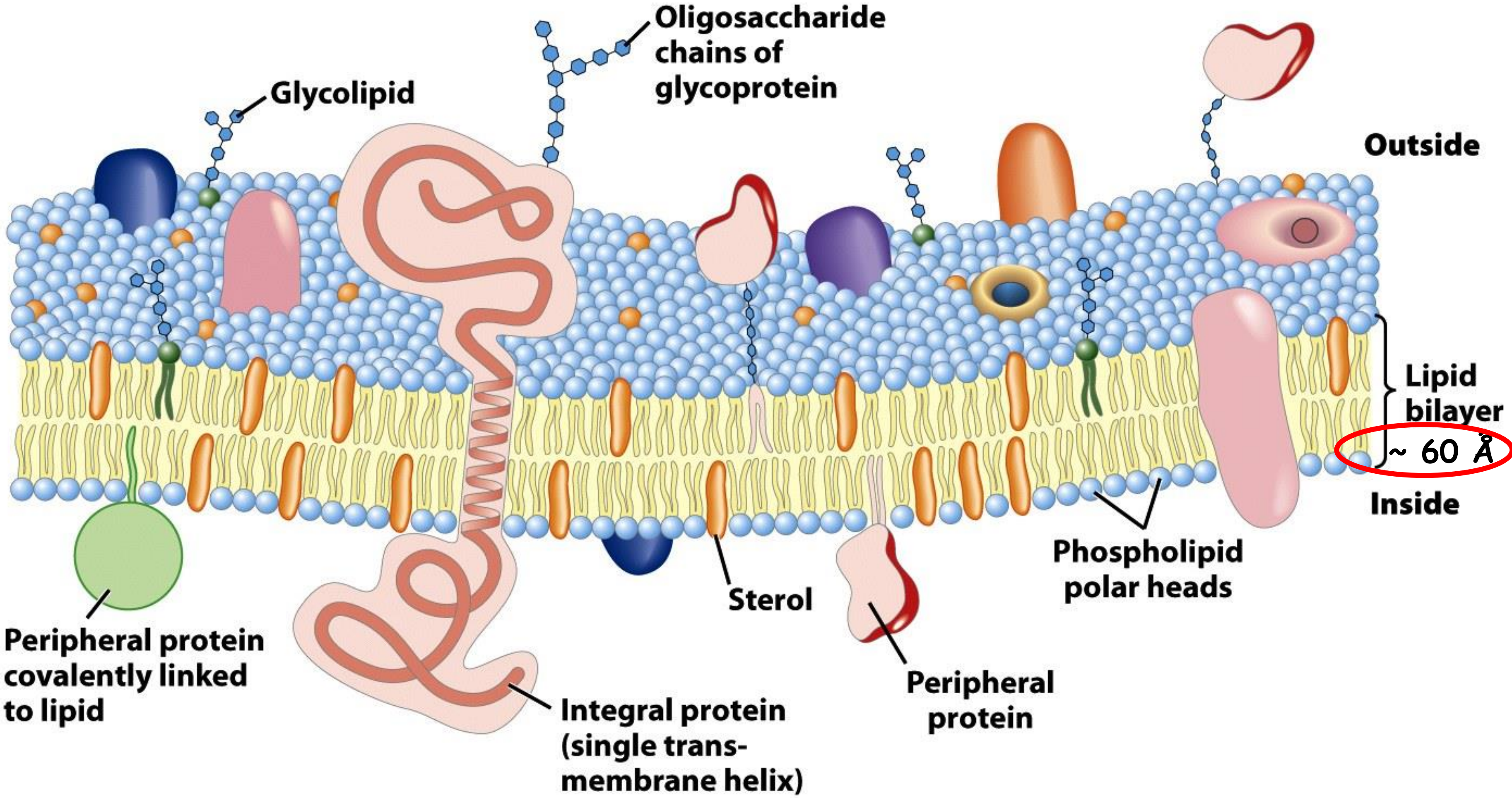

Figure 11-3
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Membranes are:

1. Non covalent assemblies
2. Asymmetric
3. Fluid mosaic structures
4. Electrically polarized, negative inside(-60 mV)

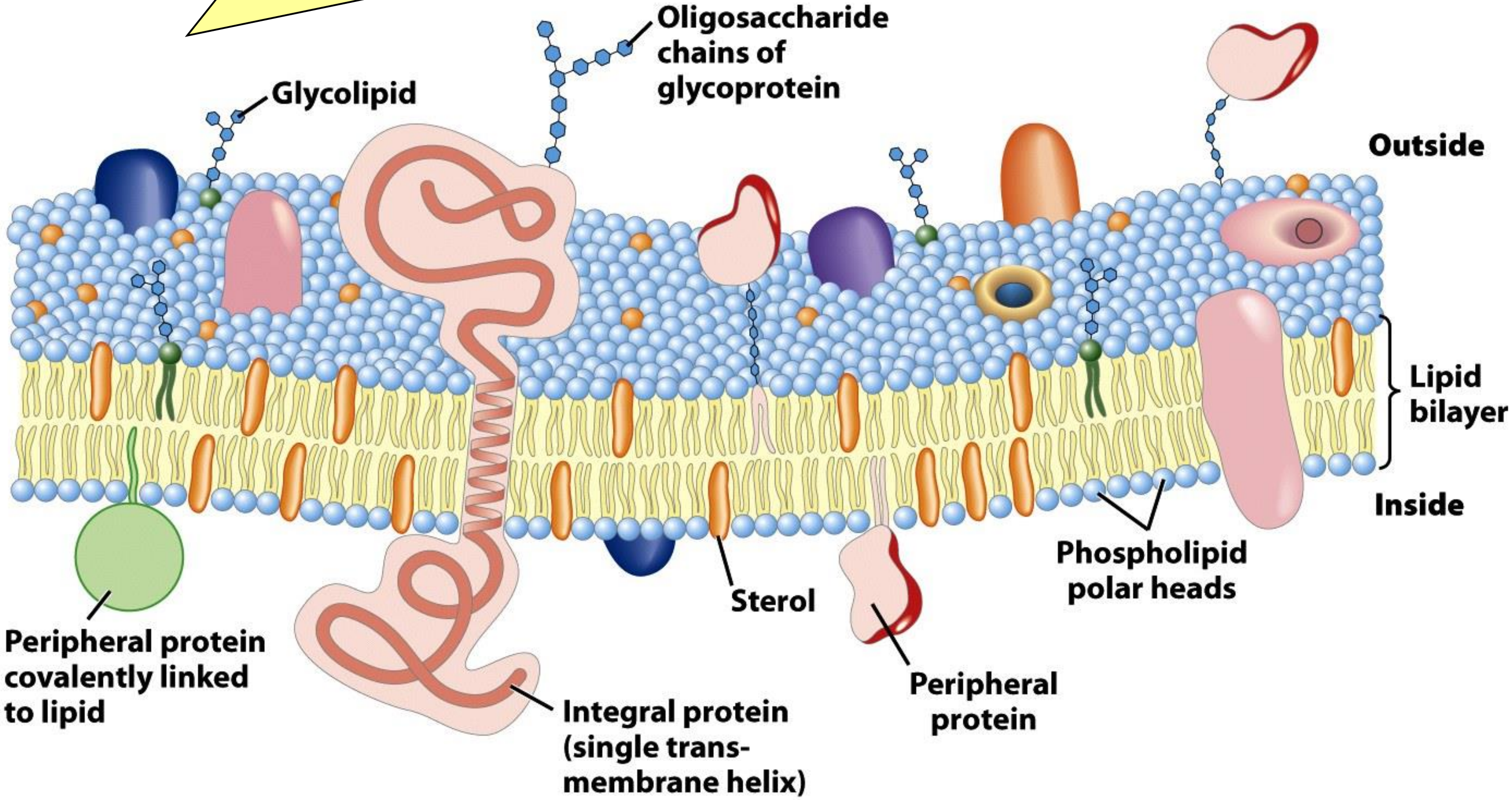

Figure 11-3
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LIPIDS

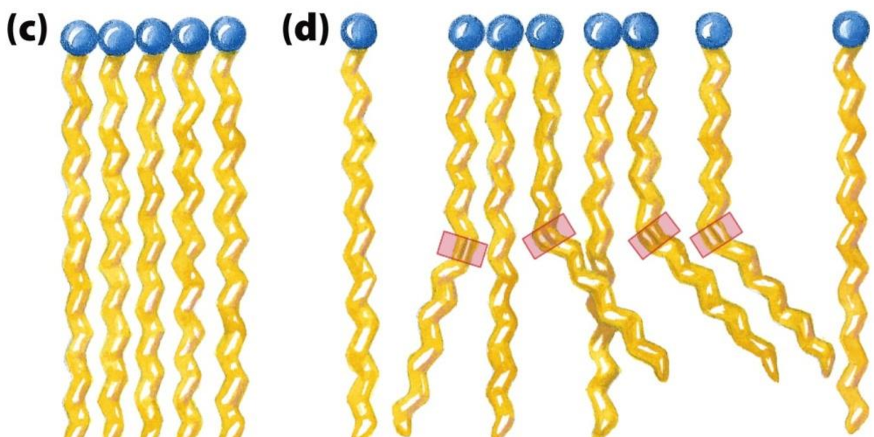
They are substances of biological origin that are soluble in organic solvents such as chlorophorm and methanol, but are only sparingly soluble, if at all, in water.

The membrane lipids, as well as the fat lipids are derivatives of Fatty Acids

FATTY ACIDS

are carboxylic acids with long hydrocarbon side groups

Hydrophilic "heads"



Saturated fatty acids

Mixture of saturated and unsaturated fatty acids

Figure 10-2cd
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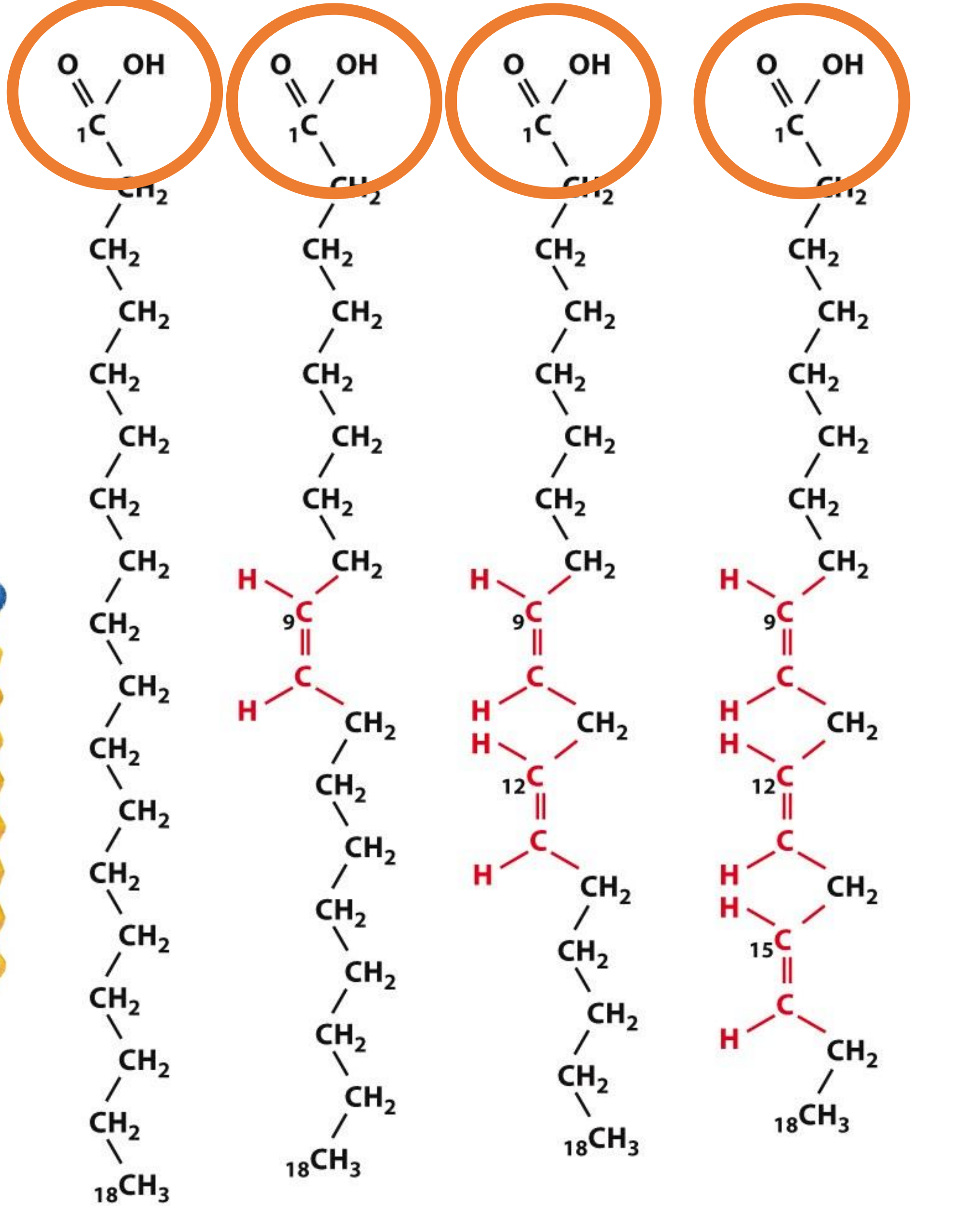
Hydrophobic "tails"



Stearic acid | Oleic acid | Linoleic acid | α-Linolenic acid

Figure 12-1

Physical properties of fatty acids

TABLE 10–1 Some Naturally Occurring Fatty Acids: Structure, Properties, and Nomenclature

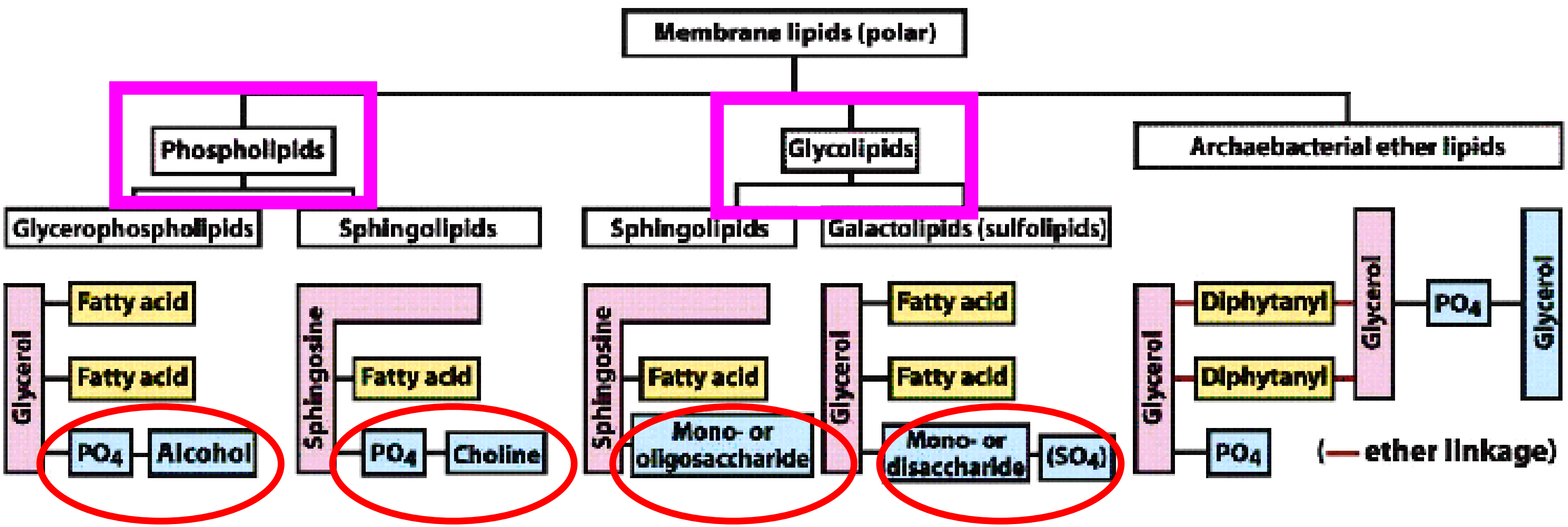
Carbon skeleton	Structure*	Systematic name†	Common name (derivation)	Melting point (°C)	Solubility at 30 °C (mg/g solvent)	
					Water	Benzene
12:0	$CH_3(CH_2)_{10}COOH$	*n*-Dodecanoic acid	Lauric acid (Latin *laurus*, "laurel plant")	44.2	0.063	2,600
14:0	$CH_3(CH_2)_{12}COOH$	*n*-Tetradecanoic acid	Myristic acid (Latin *Myristica*, nutmeg genus)	53.9	0.024	874
16:0	$CH_3(CH_2)_{14}COOH$	*n*-Hexadecanoic acid	Palmitic acid (Latin *palma*, "palm tree")	63.1	0.0083	348
18:0	$CH_3(CH_2)_{16}COOH$	*n*-Octadecanoic acid	Stearic acid (Greek *stear*, "hard fat")	69.6	0.0034	124
20:0	$CH_3(CH_2)_{18}COOH$	*n*-Eicosanoic acid	Arachidic acid (Latin *Arachis*, legume genus)	76.5		
24:0	$CH_3(CH_2)_{22}COOH$	*n*-Tetracosanoic acid	Lignoceric acid (Latin *lignum*, "wood" + *cera*, "wax")	86.0		
16:1(Δ^9)	$CH_3(CH_2)_5CH{=}CH(CH_2)_7COOH$	*cis*-9-Hexadecenoic acid	Palmitoleic acid	1 to –0.5		
18:1(Δ^9)	$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$	*cis*-9-Octadecenoic acid	Oleic acid (Latin *oleum*, "oil")	13.4		
18:2($\Delta^{9,12}$)	$CH_3(CH_2)_4CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$	*cis*-,*cis*-9,12-Octadecadienoic acid	Linoleic acid (Greek *linon*, "flax")	1–5		
18:3($\Delta^{9,12,15}$)	$CH_3CH_2CH{=}CHCH_2CH{=}CHCH_2CH{=}CH(CH_2)_7COOH$	*cis*-,*cis*-,*cis*-9,12,15-Octadecatrienoic acid	α-Linolenic acid	–11		
20:4($\Delta^{5,8,11,14}$)	$CH_3(CH_2)_4CH{=}CHCH_2CH{=}CHCH_2CH{=}CHCH_2CH{=}CH(CH_2)_3COOH$	*cis*-,*cis*-,*cis*-,*cis*-5,8,11,14-Icosatetraenoic acid	Arachidonic acid	–49.5		

*All acids are shown in their nonionized form. At pH 7, all free fatty acids have an ionized carboxylate. Note that numbering of carbon atoms begins at the carboxyl carbon.

†The prefix *n*- indicates the "normal" unbranched structure. For instance, "dodecanoic" simply indicates 12 carbon atoms, which could be arranged in a variety of branched forms; "*n*-dodecanoic" specifies the linear, unbranched form. For unsaturated fatty acids, the configuration of each double bond is indicated; in biological fatty acids the configuration is almost always cis.

Polar lipids

Membane lipids are amphipatic: one end of the molecule is hydrophobic, the other hydrophilic



1. Phospholipids: Glycerophospholipids

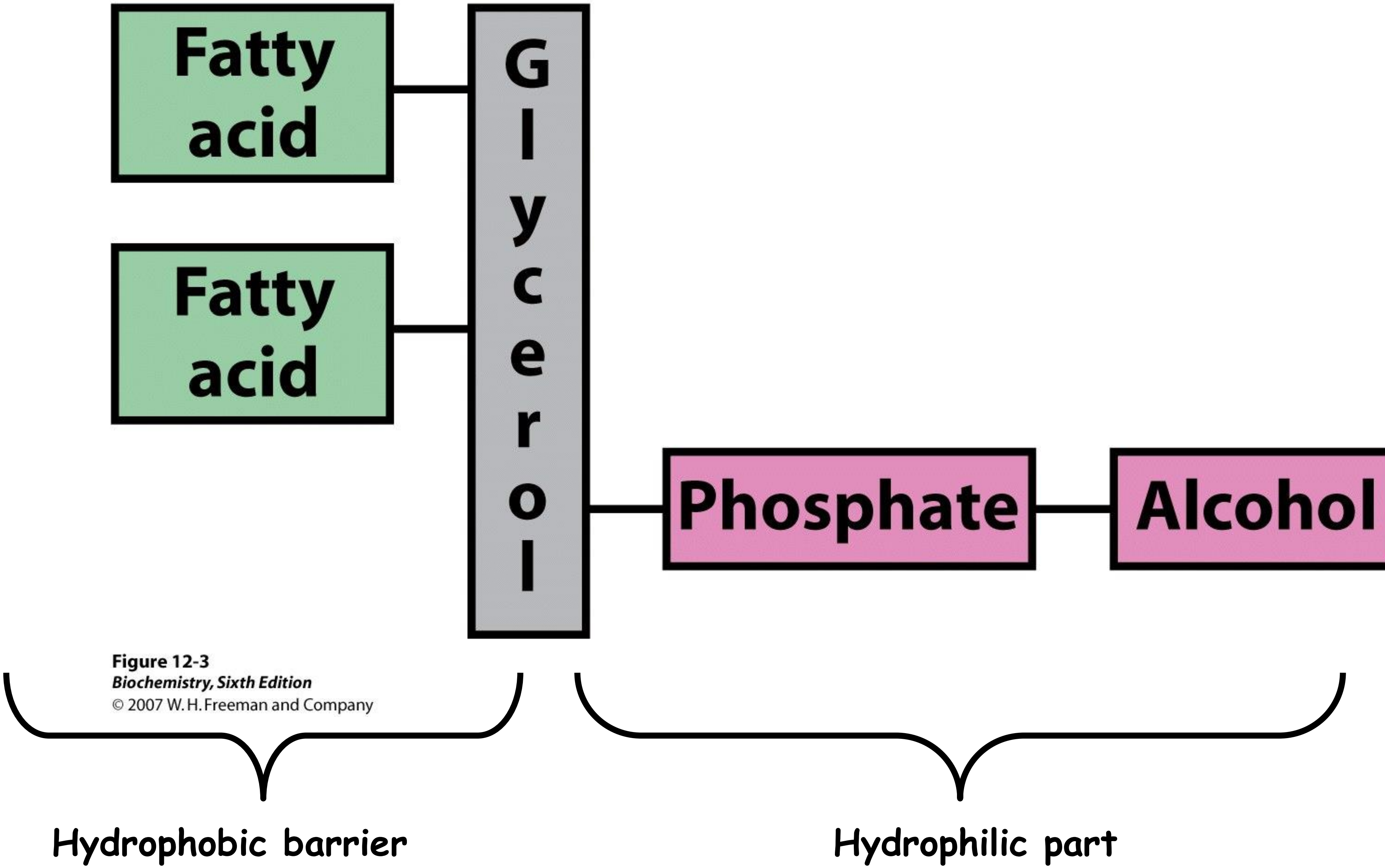


Figure 12-3
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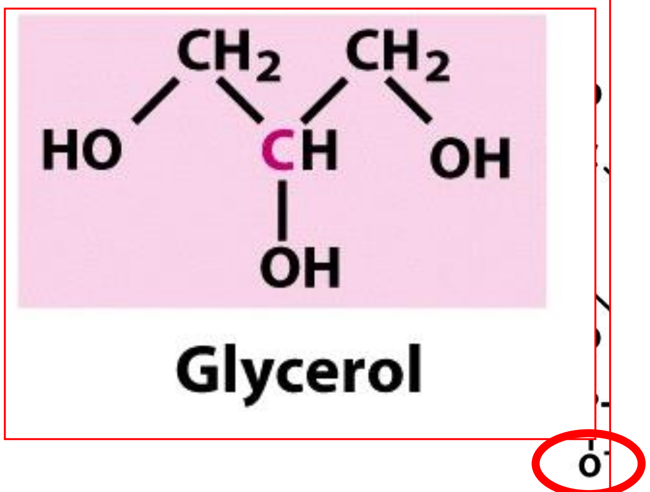
PHOSPHOGLYCERIDE
or
GLYCEROPHOSPHOLIPID



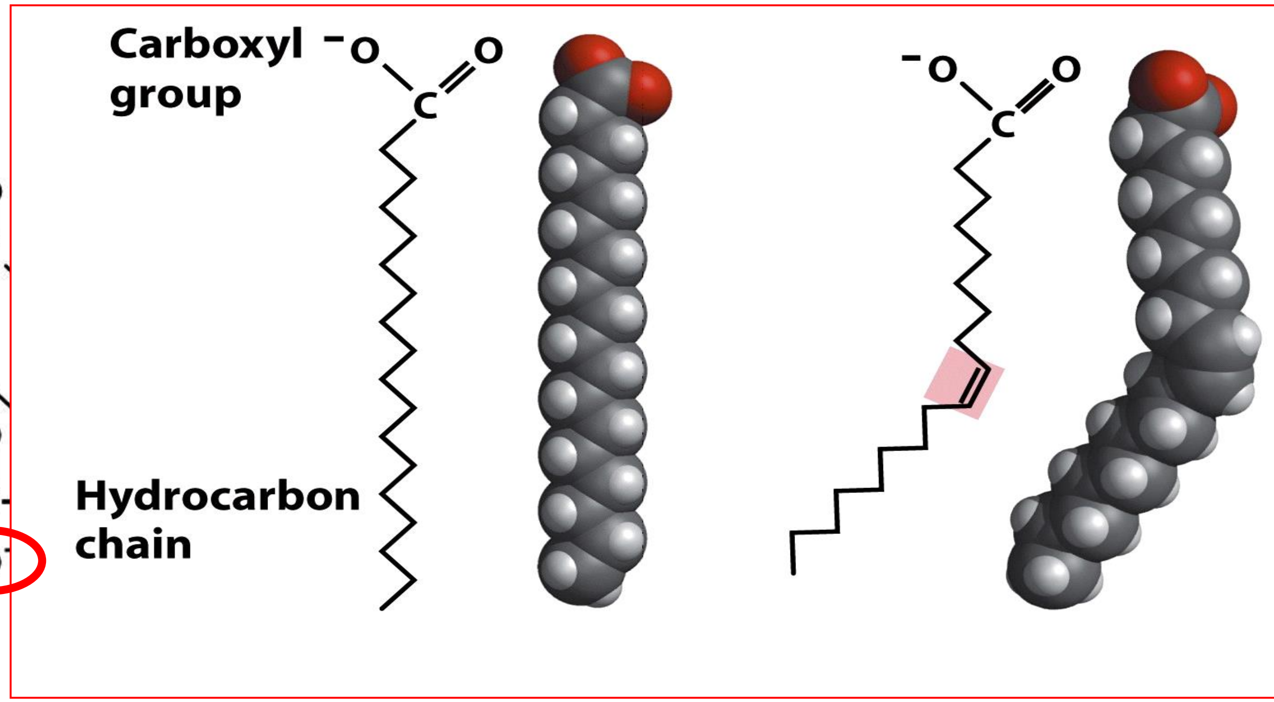


Figure 10-9 part 1
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PHOSPHATIDIC ACID IS A PHOSPHOMONOESTER, AND REPRESENTS THE PARENT COMPOUND

PHOSPHOGLYCERIDE
or
GLYCEROPHOSPHOLIPID

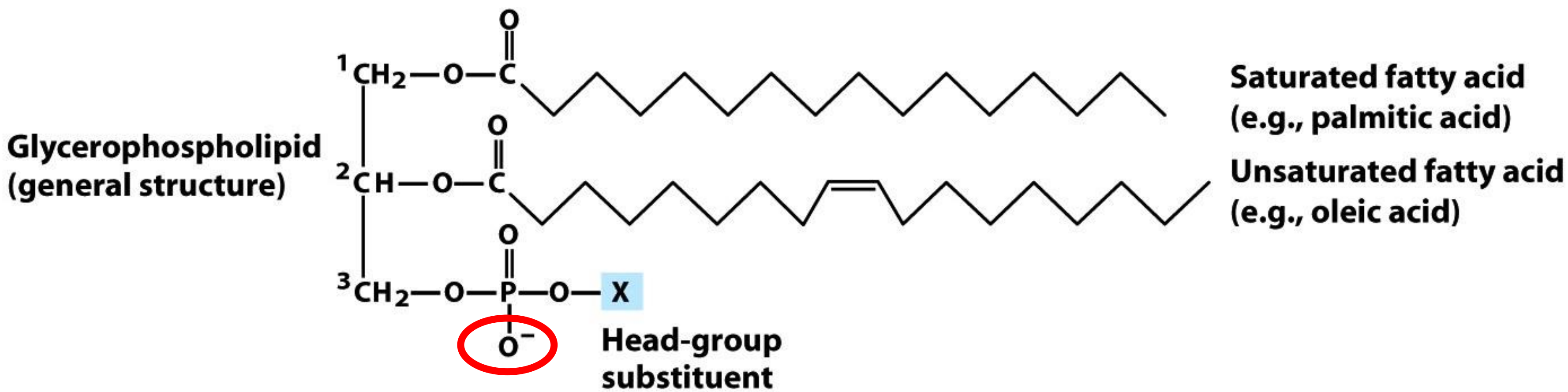

Figure 10-9 part 1
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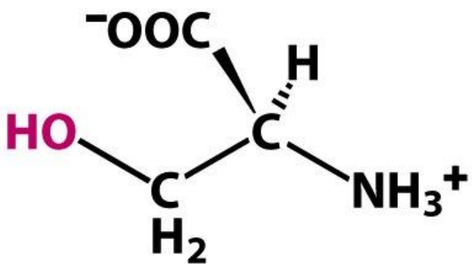
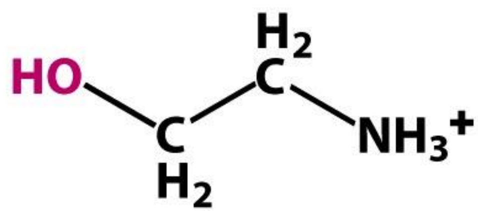
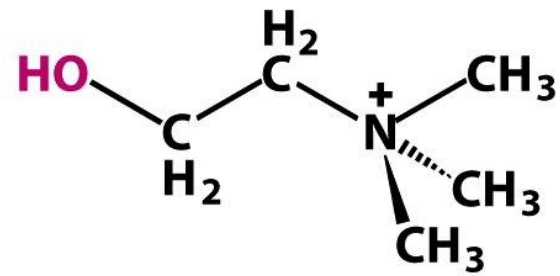
-X molecules ester bound to P



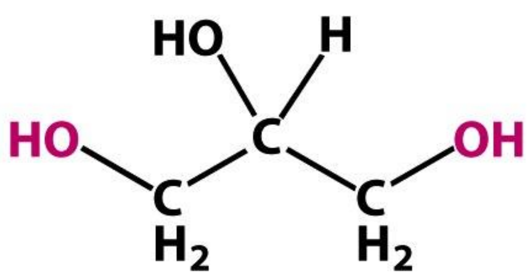
Serine



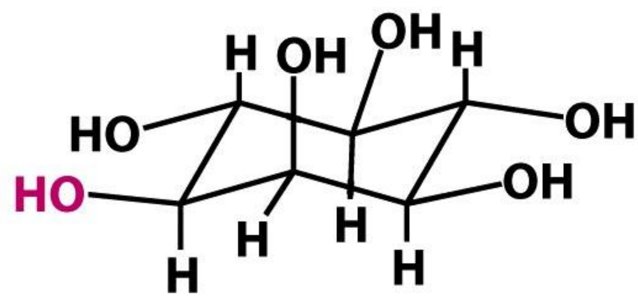
Ethanolamine



Choline



Glycerol



Inositol

Unnumbered figure pg 330
Biochemistry, Sixth Edition

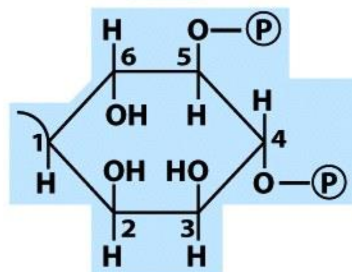
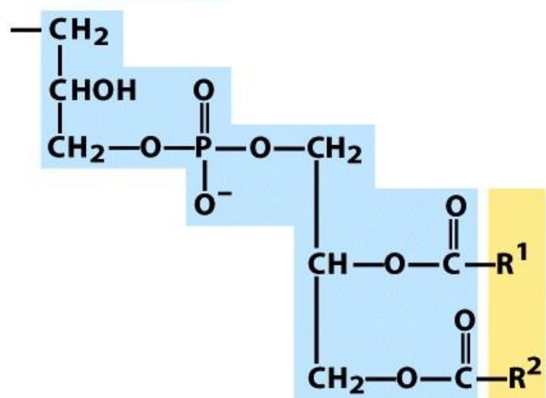
Name of glycerophospholipid	Name of X	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—	$-H$	– 1
Phosphatidylethanolamine	Ethanolamine	$-CH_2-CH_2-\overset{+}{N}H_3$	0
Phosphatidylcholine	Choline	$-CH_2-CH_2-\overset{+}{N}(CH_3)_3$	0
Phosphatidylserine	Serine	$-CH_2-CH(COO^-)-\overset{+}{N}H_3$	– 1
Phosphatidylglycerol	Glycerol	$-CH_2-CH(OH)-CH_2-OH$	– 1
Phosphatidylinositol 4,5-bisphosphate	*myo*-Inositol 4,5-bisphosphate		– 4
Cardiolipin	Phosphatidyl-glycerol		– 2

Figure 10-9 part 2
Lehninger Principles of Biochemistry, Fifth Edition

GLYCEROPHOSPOLIPIDS WITH ETER-LINK FATTY ACIDS

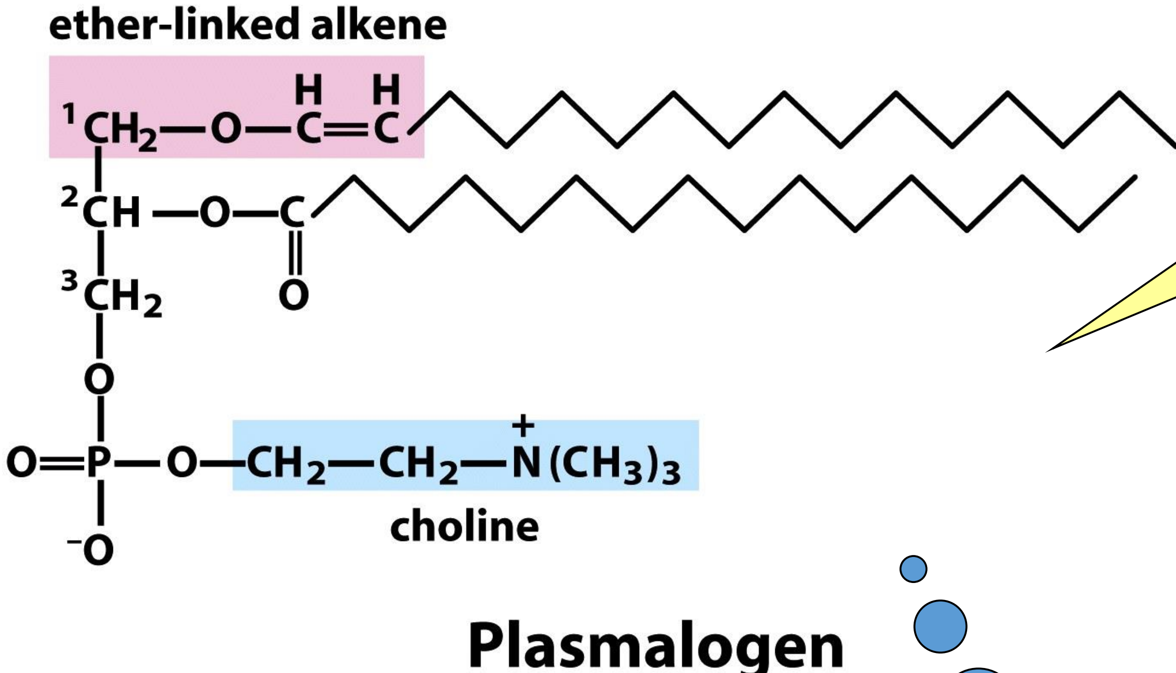
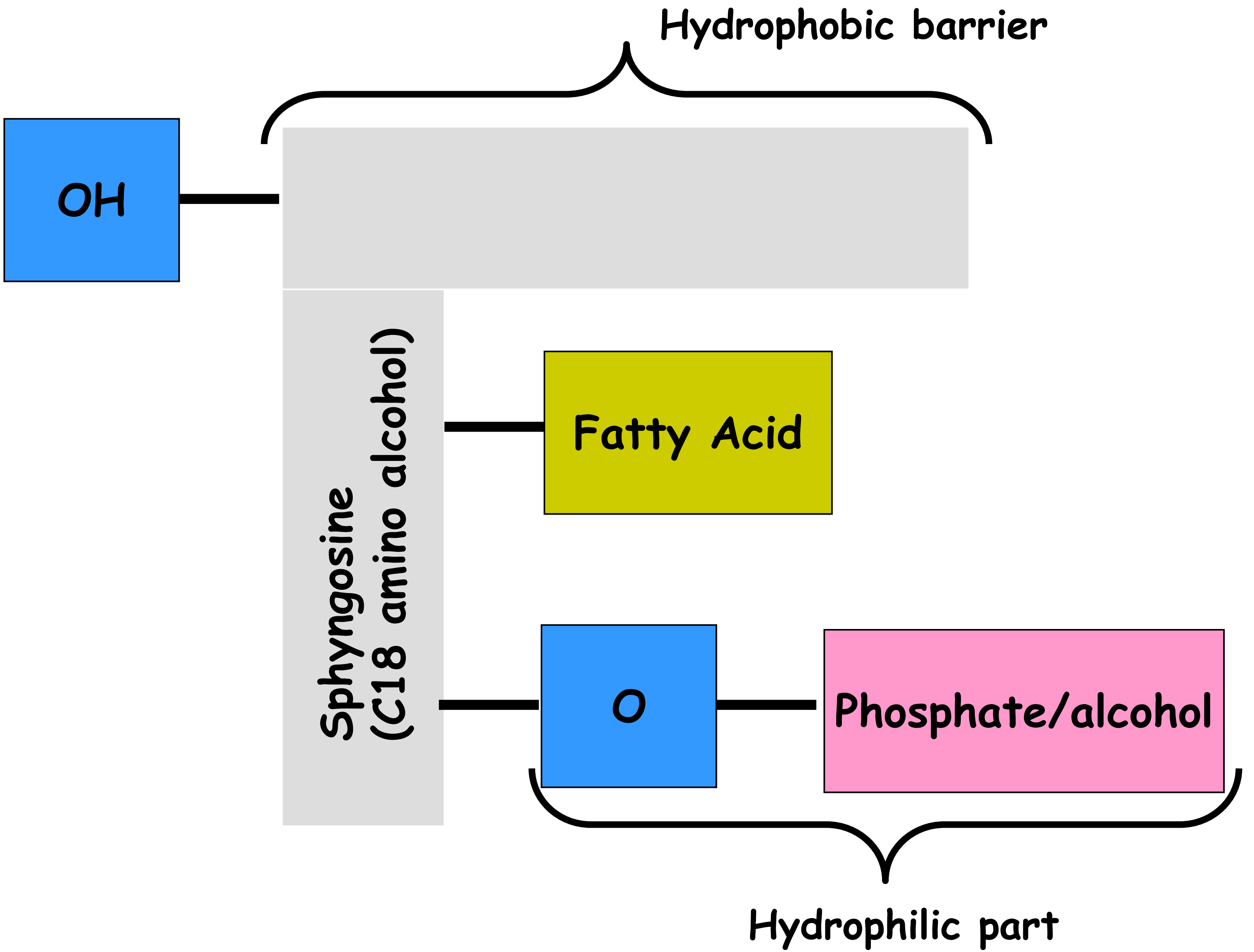


Figure 10-10a
Lehninger Principles of Biochemistry, Fifth Edition

Particularly abundant in Hearth and Nervous Tissues.

High level of ether linked lipids had been found in cancer metastatic cells. Thus it had been hypothized a role for the lipids in the invasive properties of these cells.

2.Phospholipids
Sphingolipids



SPHINGOLIPIDS

Ceramide is the structural parent of all sphingolipids

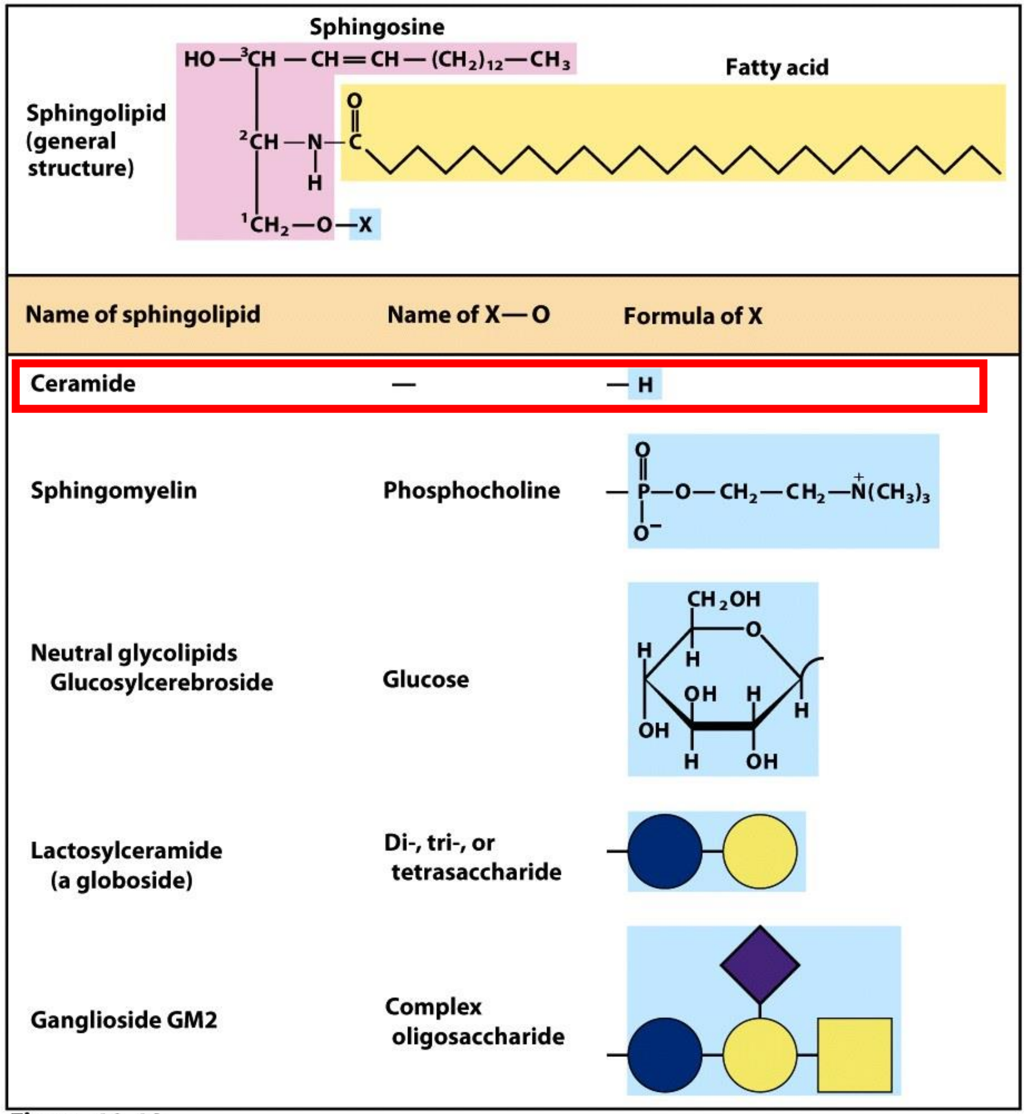


Figure 10-13
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Sphingomyelins are sphingophospholipids that are classified, together with glycerophospholipids, as PHOSPHOLIPIDS

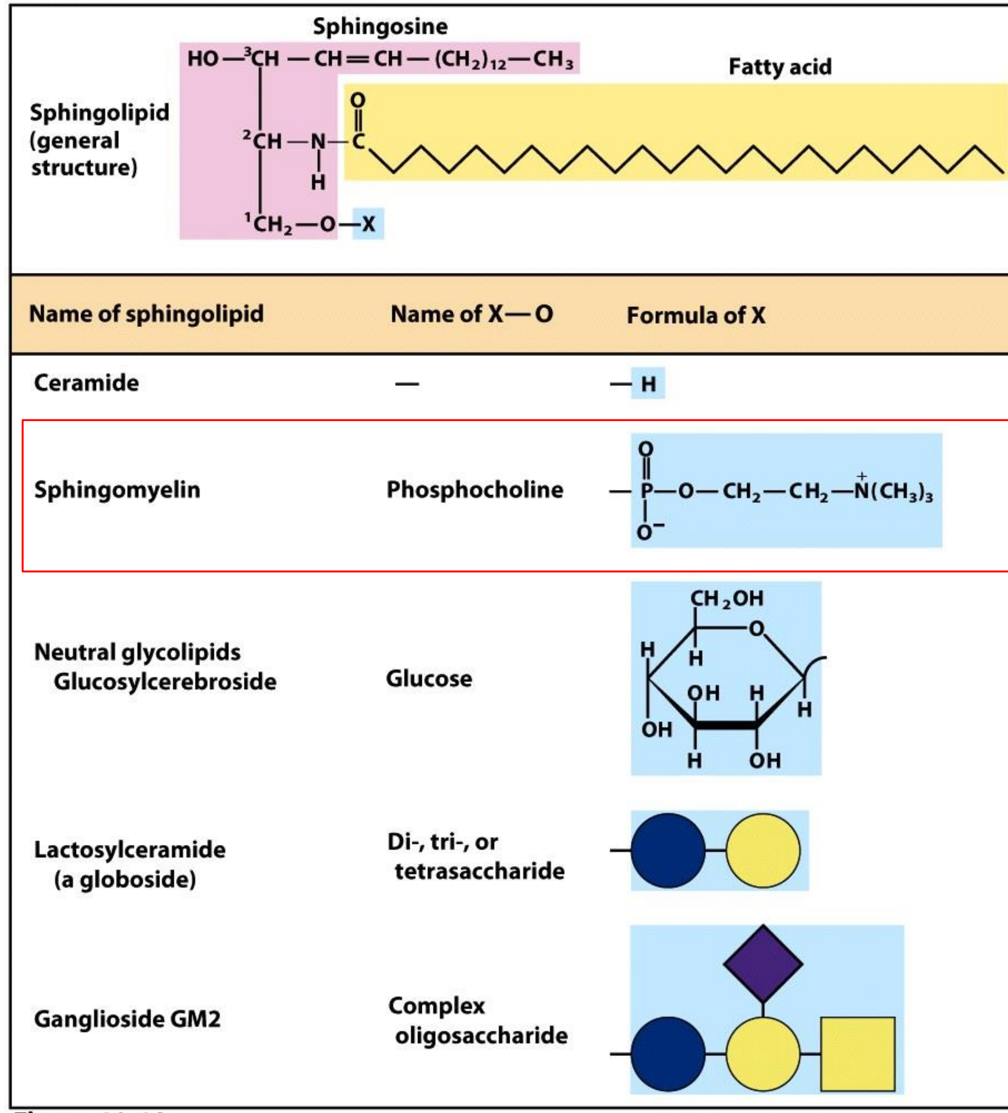

Figure 10-13

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Glycosphingolipids or Sphingoglycolipids

They contain a sugar molecule linked by a β-glycosidic bond to C1 OH group of a ceramide; <u>they lack phosphate.</u>

Galactocerebroside

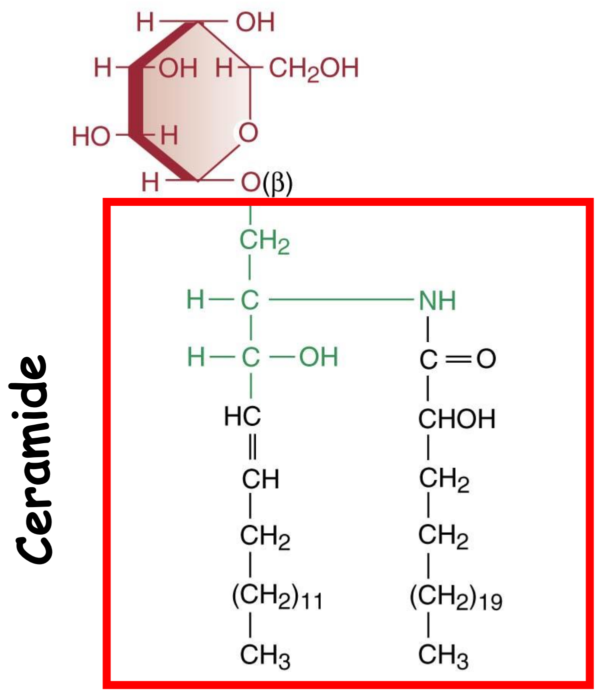

Figure 12.14. Structure of a galactocerebroside containing a C_{24} fatty acid.

Sulfatide

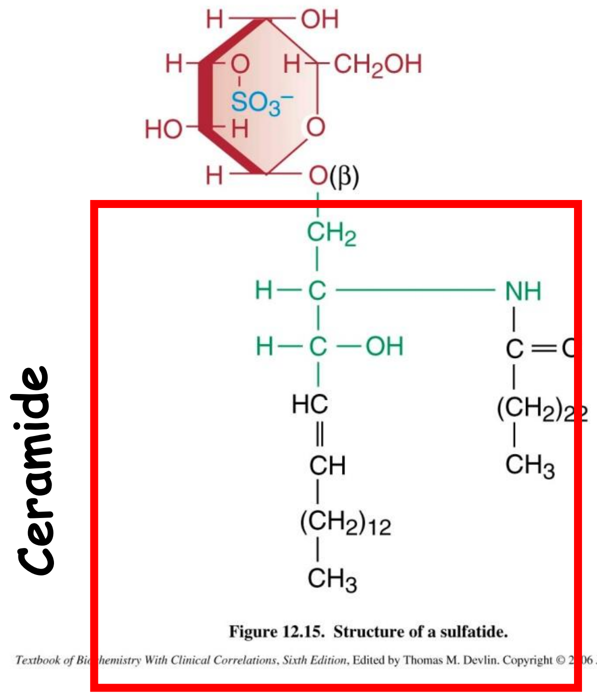

Figure 12.15. Structure of a sulfatide.

SPHINGOLIPIDS

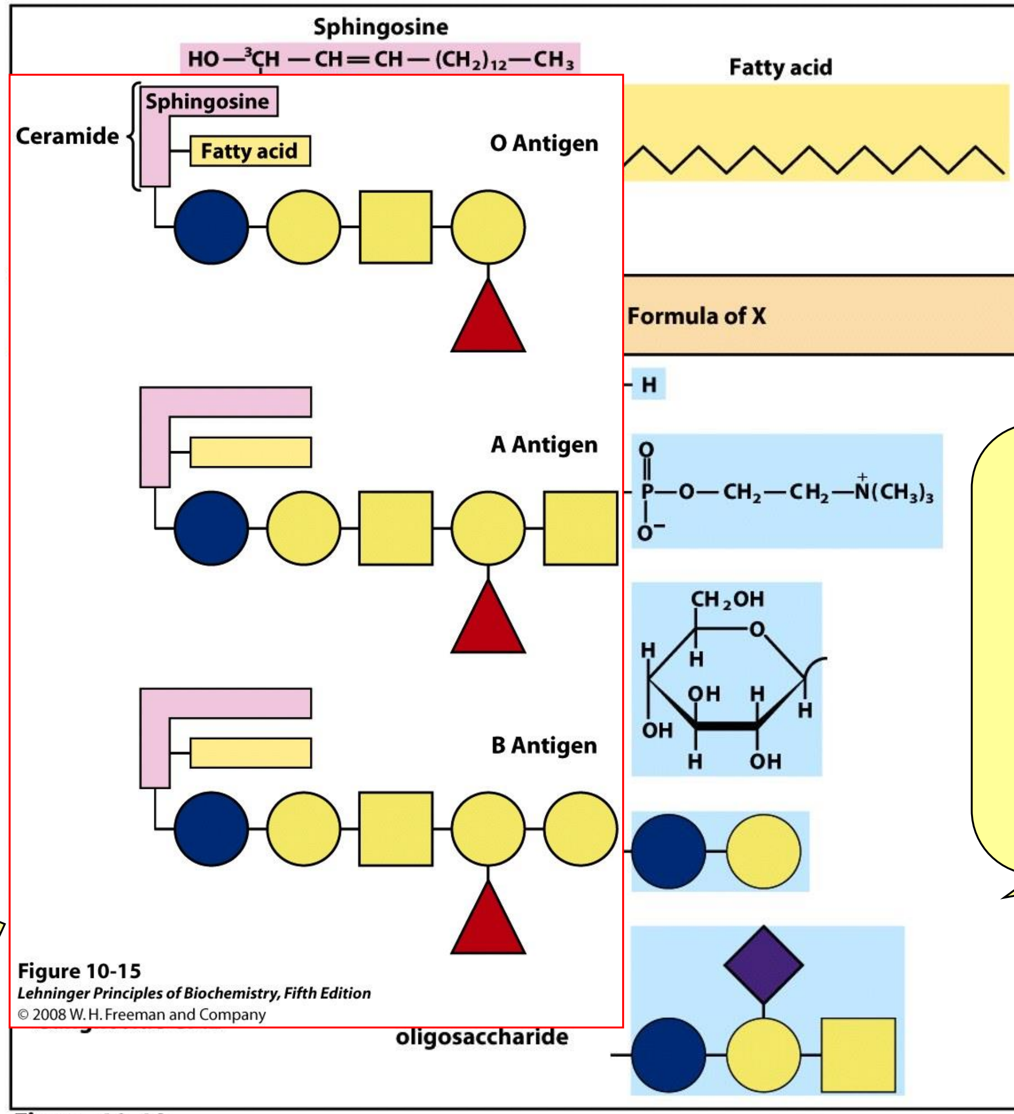

Figure 10-15
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Figure 10-13
Lehninger Principles of Biochemistry, Fifth Edition

The carbohydrates of some sphingolipids define the human Blood group.

The sugar residues are oriented toward the outer surface of the cells

Gangliosides:

are the most complex glycosphingolipids

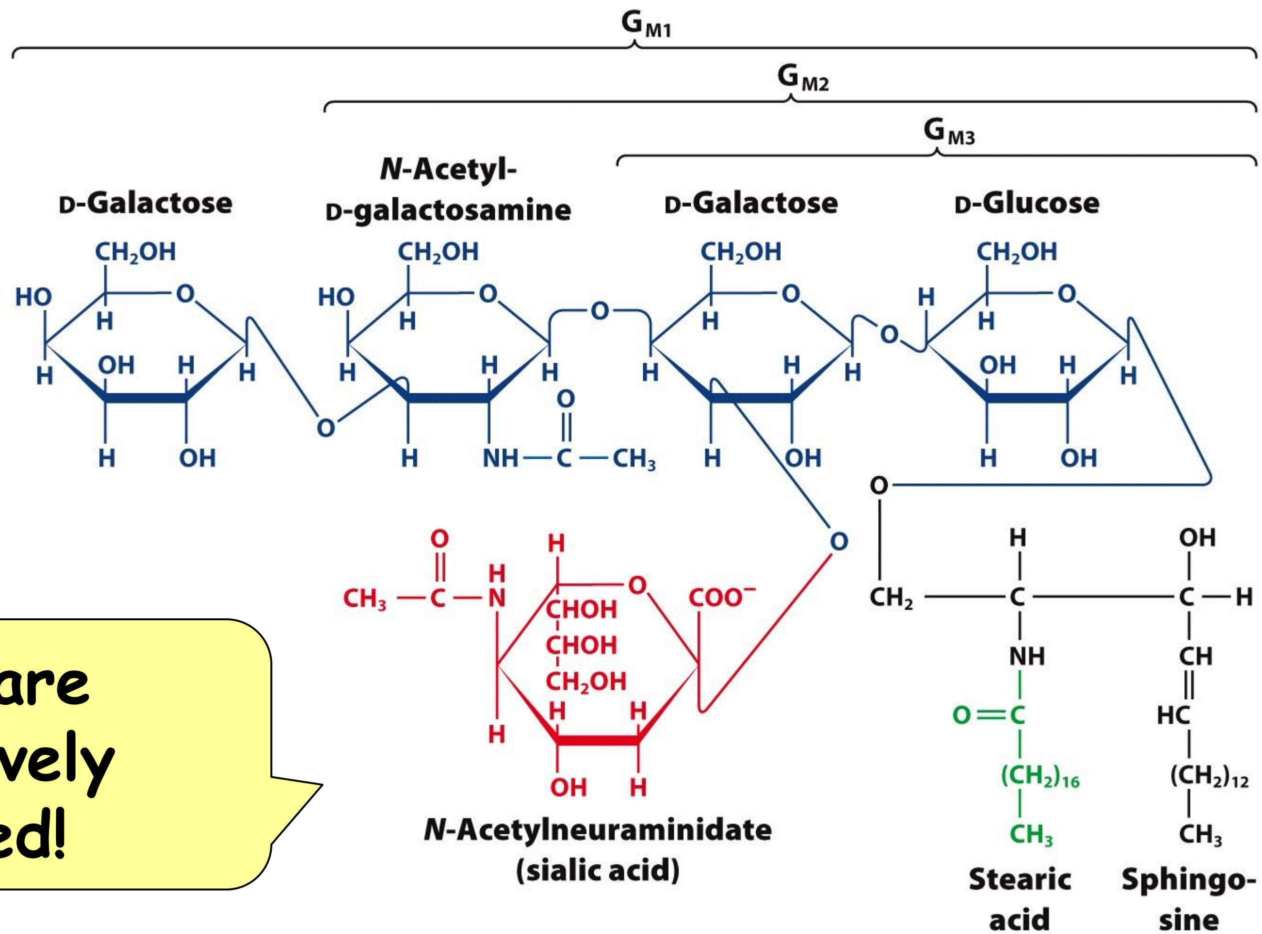

Figure 12-7a

They are negatively charged!

Membrane lipids turnover is mediated by lysosomal enzymes

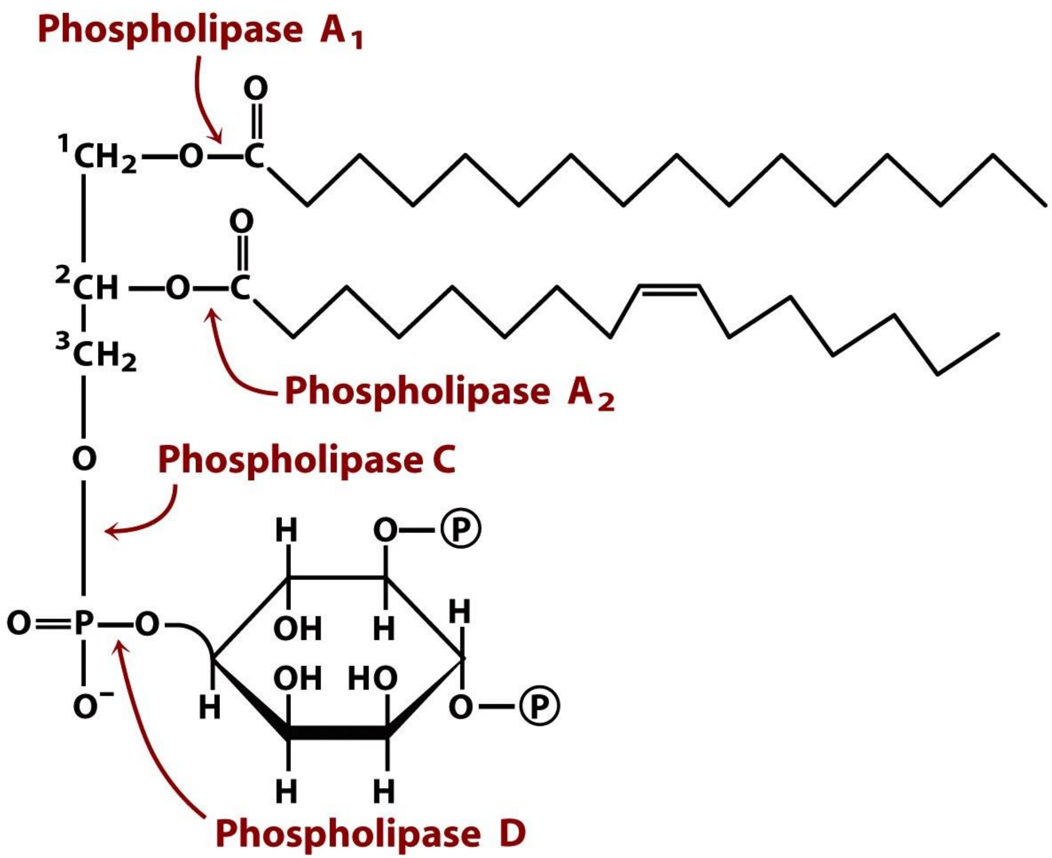

Figure 10-16
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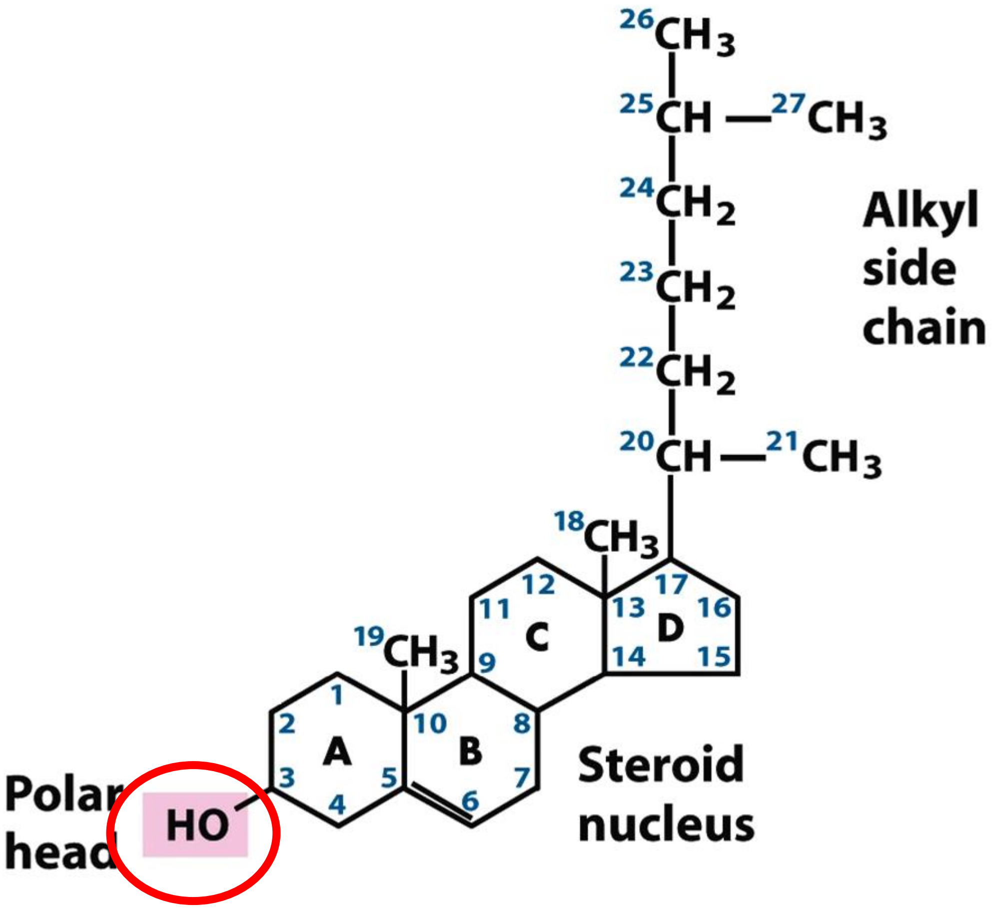
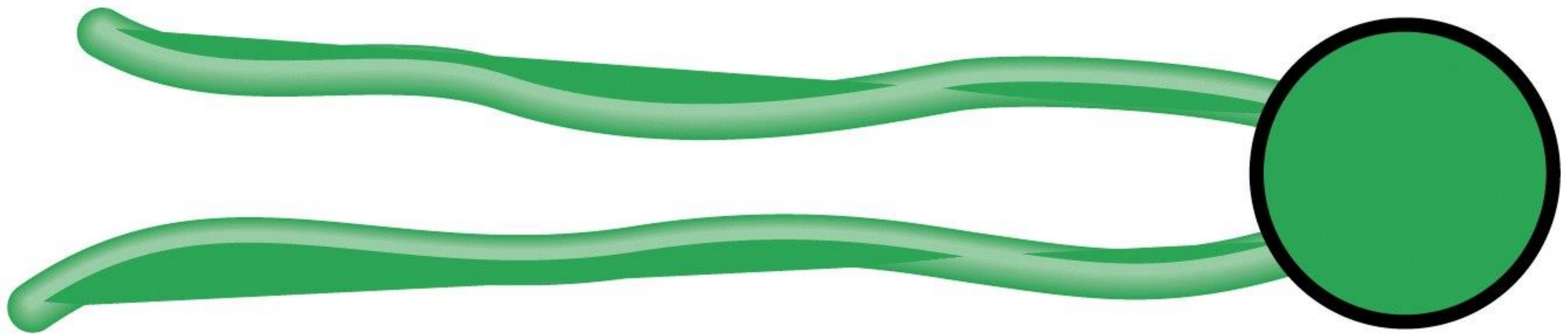
Polar lipids



CHOLESTEROL

Polar Lipids scheme



Shorthand depiction

Figure 12-8b
Biochemistry, Sixth Edition

Glycerophospholipids, sphingolipids and sterols are insoluble in water, so what is happening when they are in aqueous solution?

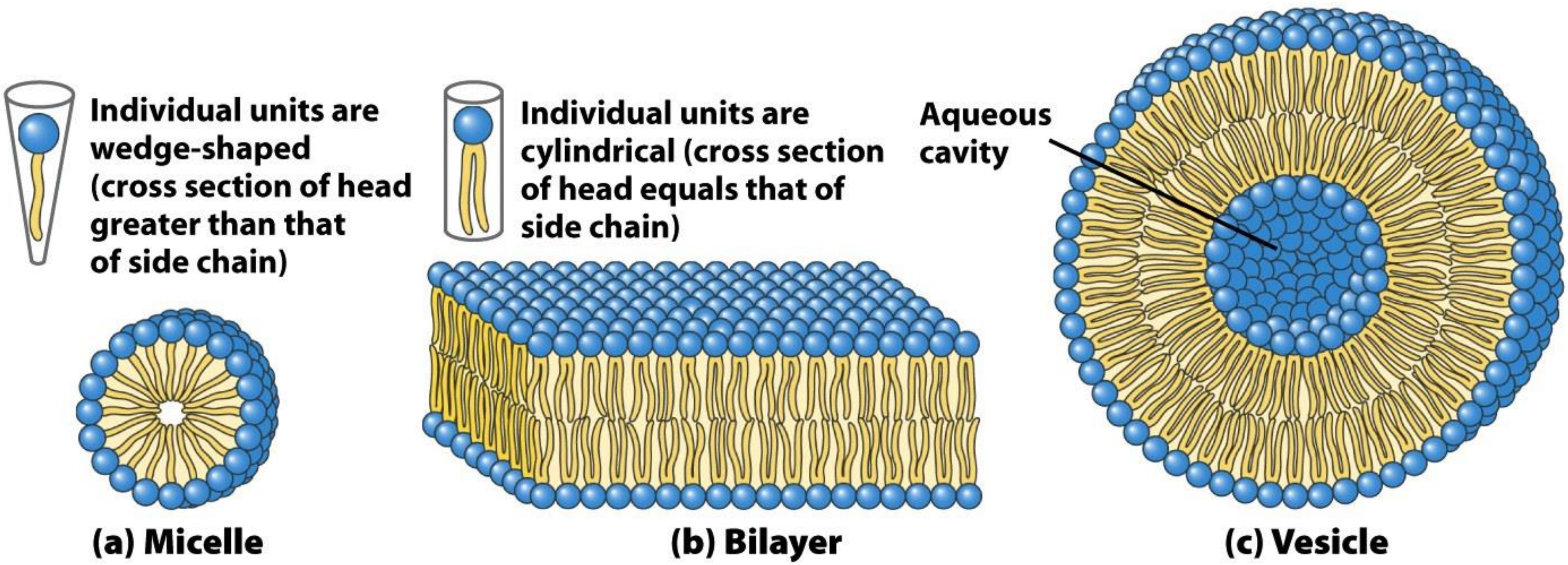


Figure 11-4
Lehninger Principles of Biochemistry, Fifth Edition

Membrane lipids are amphipatic molecules.
They have both a hydrophilic and hydrophobic moiety

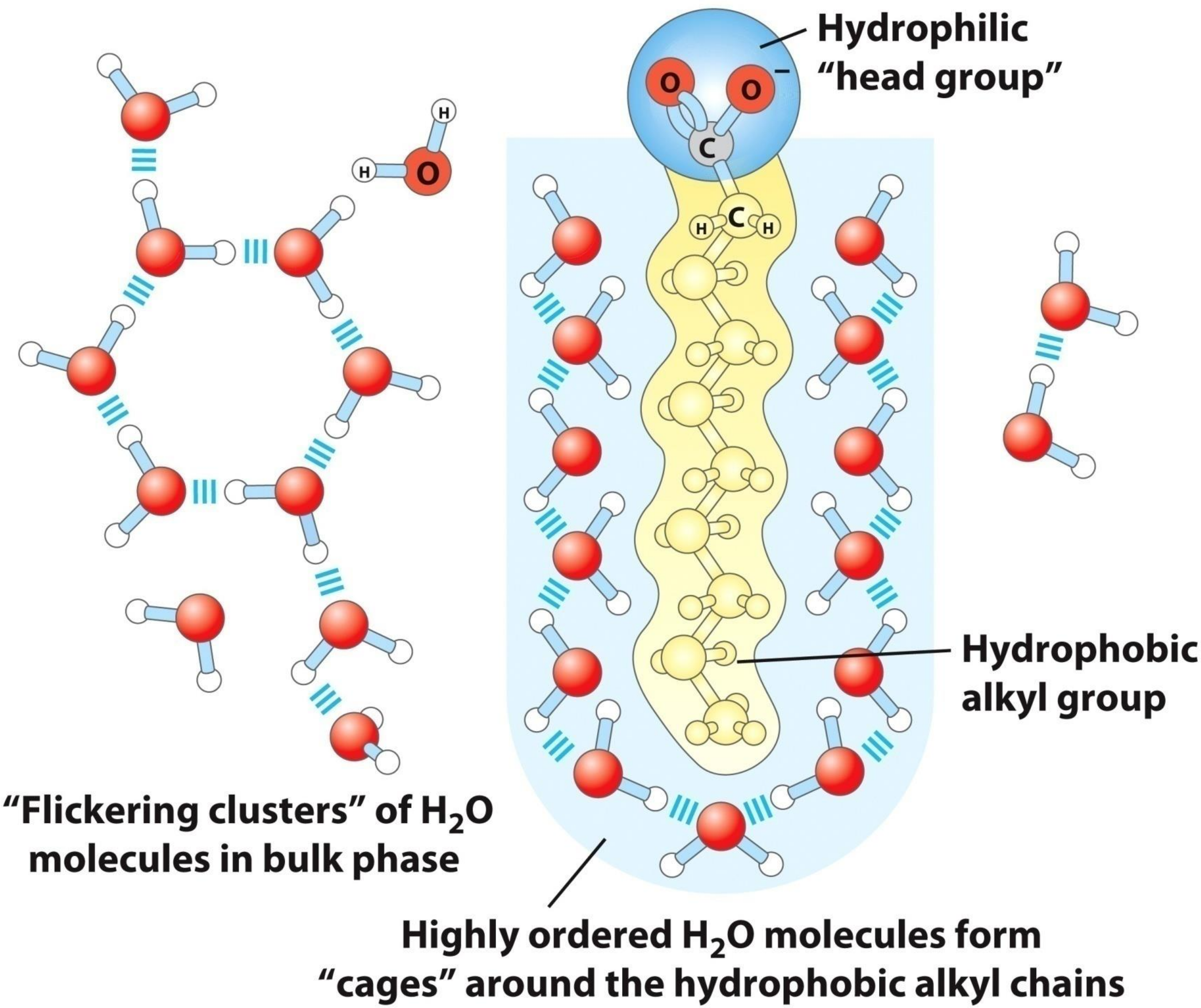
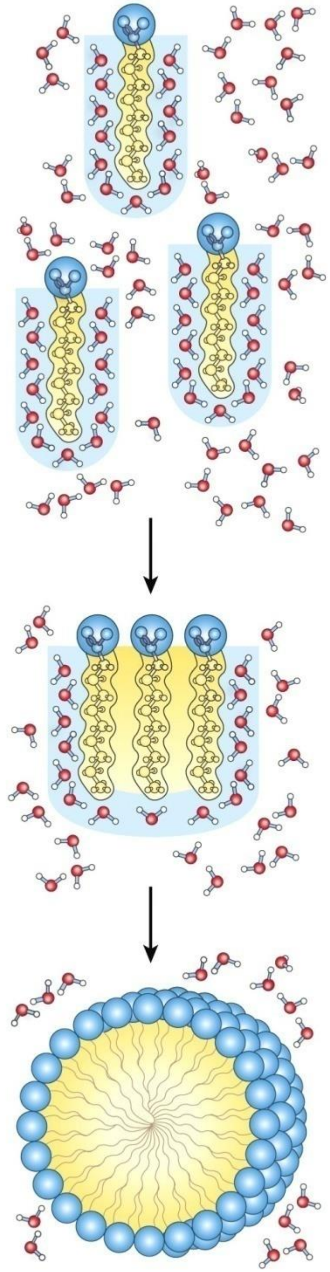


Figure 2-7a

Lehninger Principles of Biochemistry, Sixth Edition



Dispersion of lipids in H_2O

Each lipid molecule forces surrounding H_2O molecules to become highly ordered.

Clusters of lipid molecules

Only lipid portions at the edge of the cluster force the ordering of water. Fewer H_2O molecules are ordered, and entropy is increased.

Micelles

All hydrophobic groups are sequestered from water; ordered shell of H_2O molecules is minimized, and entropy is further increased.

Figure 2-7b

Lehninger Principles of Biochemistry, Sixth Edition

Glycerophospolipids, sphingolipids and sterols are insoluble in water, so what is happening when they are in aqueous solution?

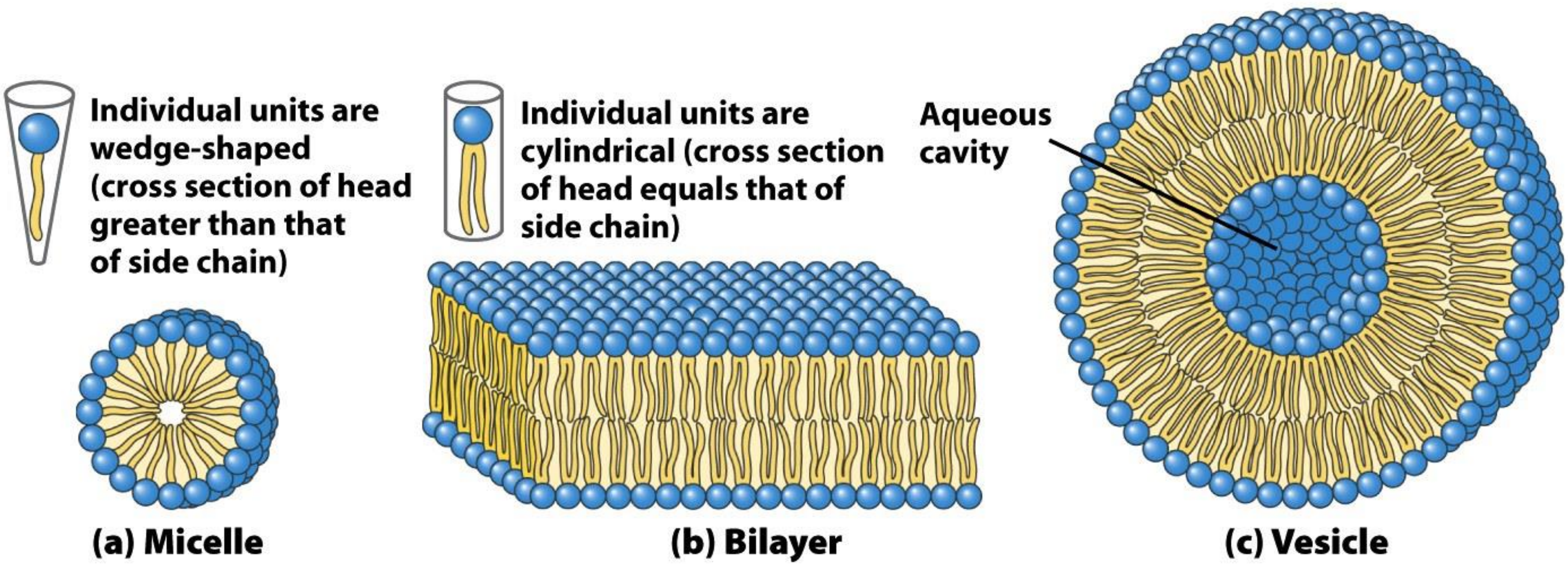
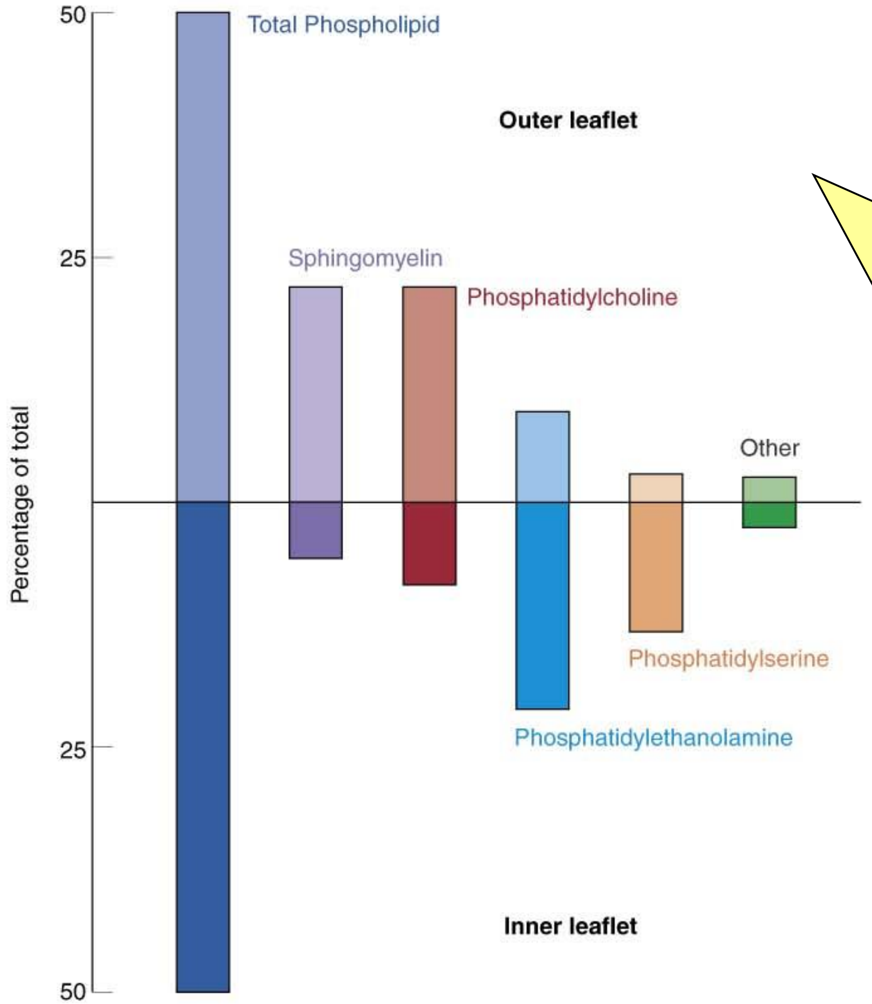


Figure 11-4
Lehninger Principles of Biochemistry, Fifth Edition

Membrane lipids are amphipatic molecules.
They have both a hydrophilic and hydrophobic moiety

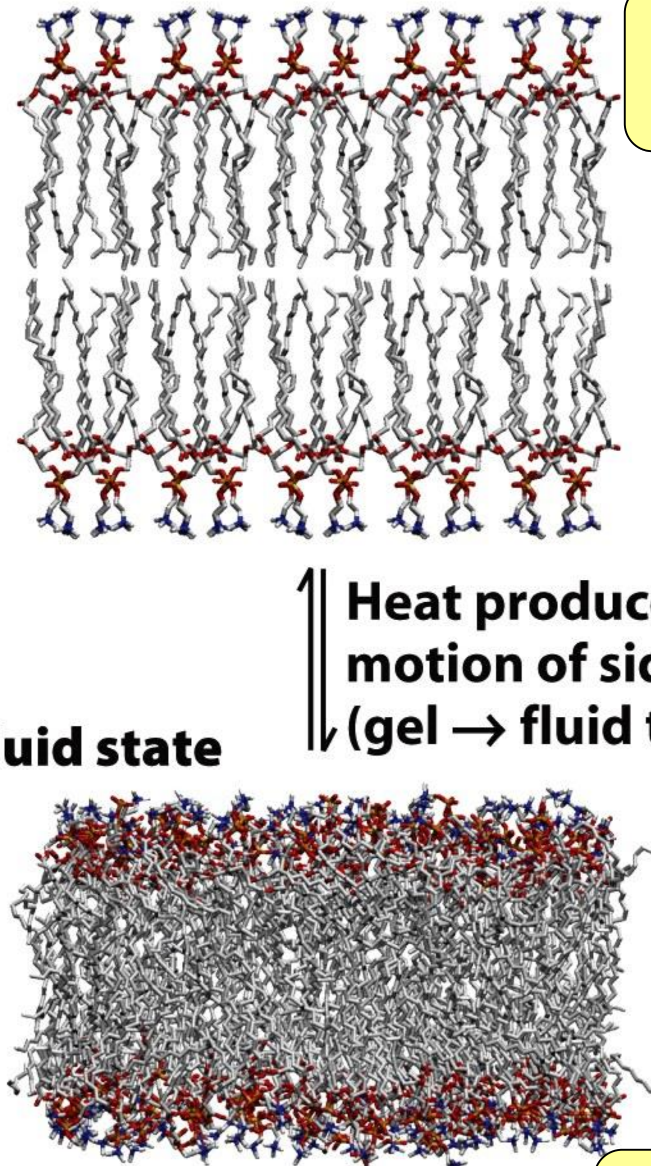


The distribution of lipids is asymmetric in plasma membrane and it is different in different cell types

Figure 12.23. Distribution of phospholipids between inner and outer layers of the human erythrocyte membrane. Data from A. J. Verkeij, R. F. A. Zwaal, B. Roelofsen, P. Comfurius, D. Kastelijn, and L. L. M. Van Deenan. The asymmetric distribution of phospholipids in the human red cell membrane. *Biochim. Biophys. Acta* 323:178, 1973. Zachowski, A. Phospholipids in animal eukaryotic membranes: Transverse symmetry and movement. *Biochem. J.* 294:1, 1993.

The biological membranes are flexible due to the presence of lipids which are not covalently anchored.

Below Physiological T



The lipid bilayer constituting the biological membranes exists in a liquid-ordered state at physiologic temperature

Above Physiological T

Figure 11-15
Lehninger Principles of Biochemistry, Fifth Edition

What regulates membrane fluidity?

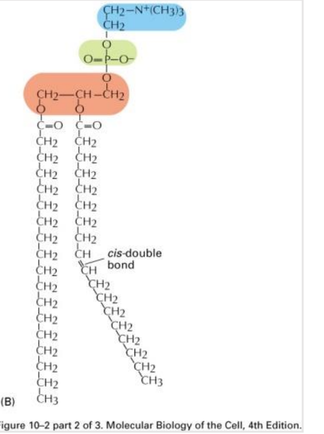


Figure 10-2 part 2 of 3. Molecular Biology of the Cell, 4th Edition.

<u>Key criterium:</u>
How densely are the lipids packed?

<u>Factors that affect lipid packing ratio:</u>

1. **length of the hydrocarbon tail**
   -lipids with shorter hydrocarbon chain interact less tightly with their neighbors
   ➡ (*membrane is <u>more</u> fluid*)

2. **degree of saturation of the hydrocarbon chain**
   -double bonds in lipid tails add "kinks", prevent tight packaging
   ➡ (*membrane is <u>more</u> fluid*)

TABLE 12.3 The melting temperature of phosphatidylcholine containing different pairs of identical fatty acid chains

		Fatty acid		
Number of carbons	Number of double bonds	Common name	Systematic name	T_m (°C)
22	0	Behenate	*n*-Docosanote	75
18	0	Stearate	*n*-Octadecanoate	58
16	0	Palmitate	*n*-Hexadecanoate	41
14	0	Myristate	*n*-Tetradecanoate	24
18	1	Oleate	*cis*-Δ^9-Octadecenoate	−22

Table 12-3
Biochemistry, Sixth Edition

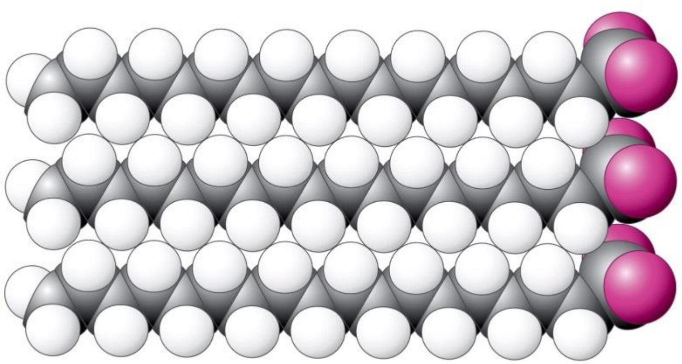


Figure 12-33a
Biochemistry, Sixth Edition

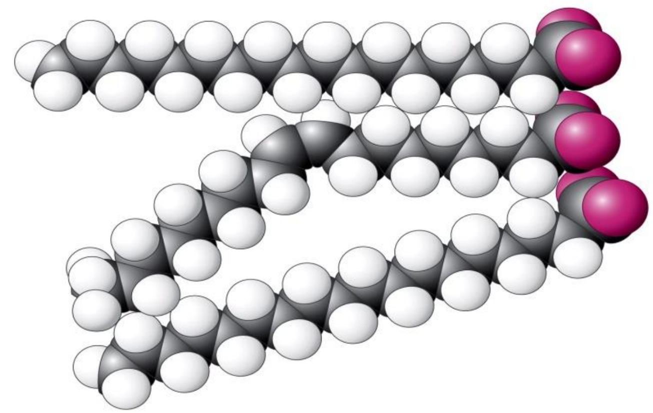


Figure 12-33b
Biochemistry, Sixth Edition

What regulates membrane fluidity? (cont.)

Factors that affect lipid packing ratio:

1. **length of the hydrocarbon tail**
   (*short tails → membrane is* *__more__* *fluid*)
2. **degree of saturation of the hydrocarbon chain**
   (*more double bonds → membrane is* *__more__* *fluid*)

3. **cholesterol content of membrane**
   -cholesterol prevents movement lipid tails ➡
   membrane *__less__* *fluid*

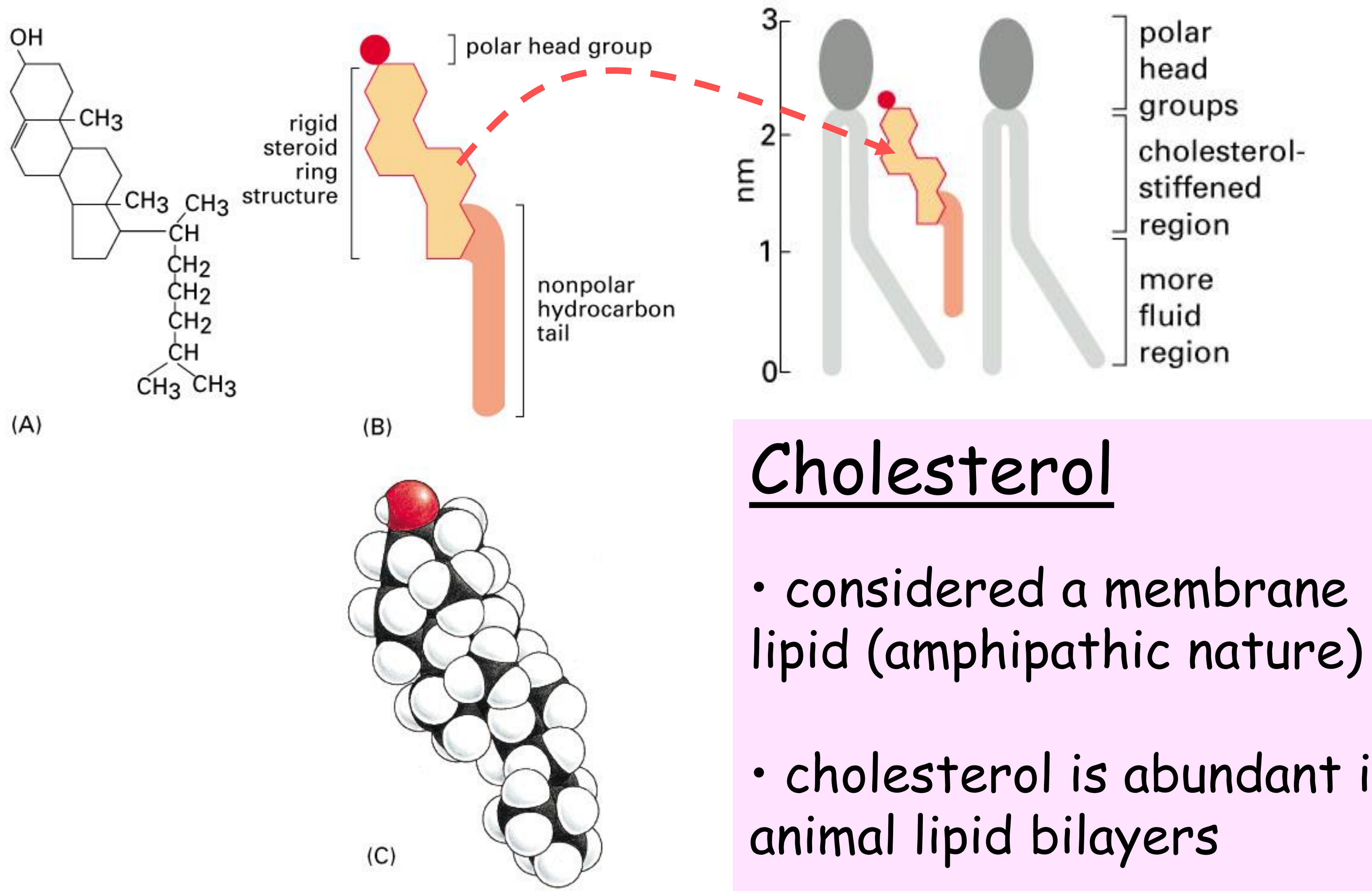


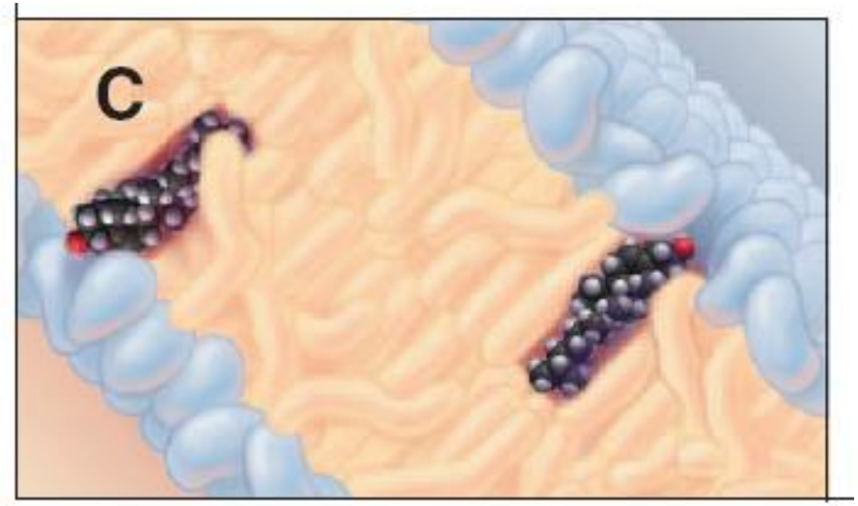
Figure 10–10. Molecular Biology of the Cell, 4th Edition.

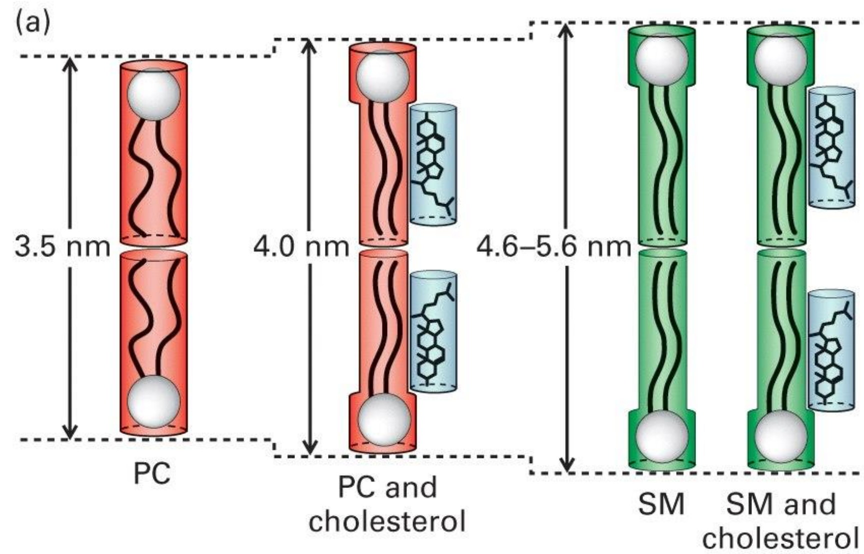
Cholesterol

• considered a membrane lipid (amphipathic nature)

• cholesterol is abundant in animal lipid bilayers

•found in both leaflets





Cholesterol affects the fluidity of the membrane

•Platelike steroid rings interact with and partially **immobilize** neighbor hydrocarbon chains; has a lipid ordering effect (for most lipids) so lipids are less fluid

•Result: cholesterol makes bilayer **less deformable** in this region, thereby **decreasing the permeability** of membrane to small water-soluble molecules

Cholesterol affects the thickness of the membrane

•Membrane thickness depends on: lipid composition, degree of backbone saturation AND cholesterol content

CAN LIPIDS MOVE IN THE MEMBRANES?

DOES THEIR MOVEMENT NEED ENERGY OR IS SPONTANEOUS?

Uncatalyzed lateral diffusion

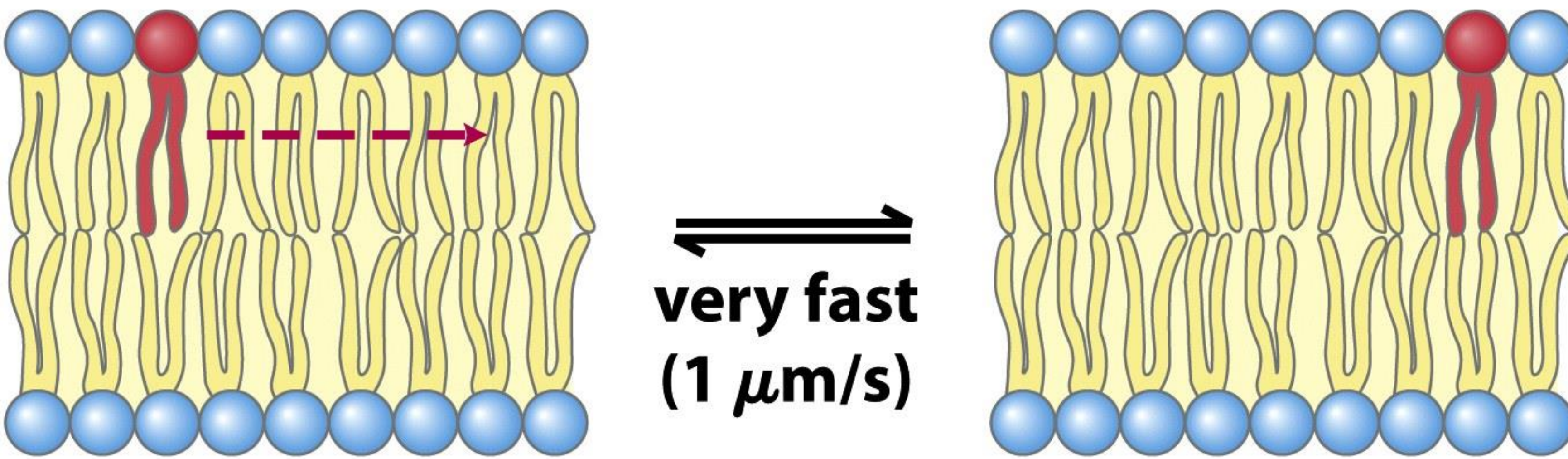


Figure 11-16b
Lehninger Principles of Biochemistry, Fifth Edition

How to “see” lateral diffusion

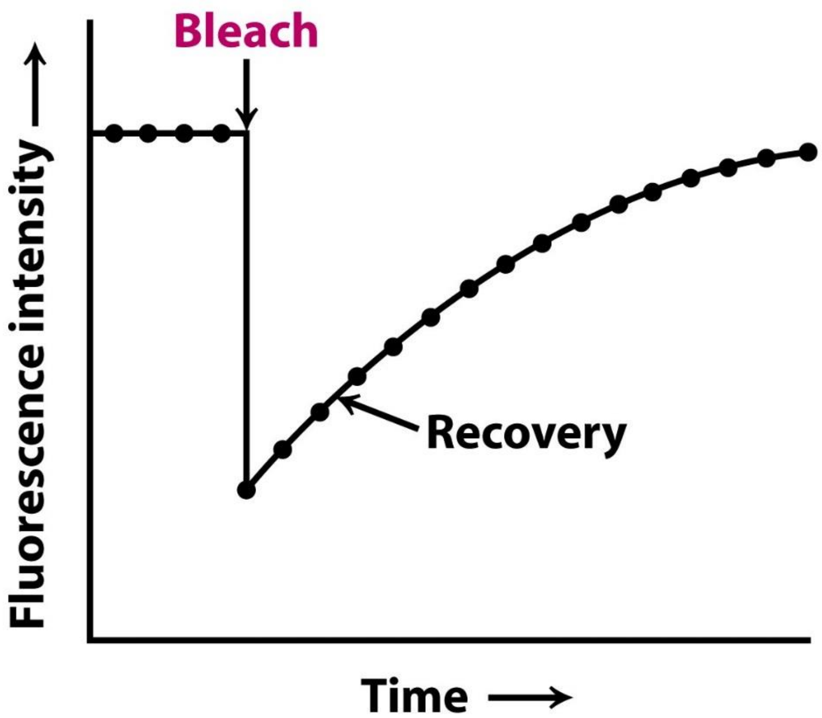


Figure 12-29d
Biochemistry, Sixth Edition

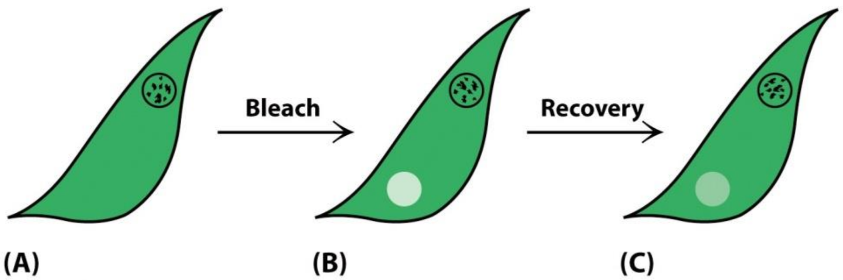


Figure 12-29abc
Biochemistry, Sixth Edition

FRAP or fluorescence recovery after photobleaching.

Uncatalyzed transbilayer ("flip-flop") diffusion

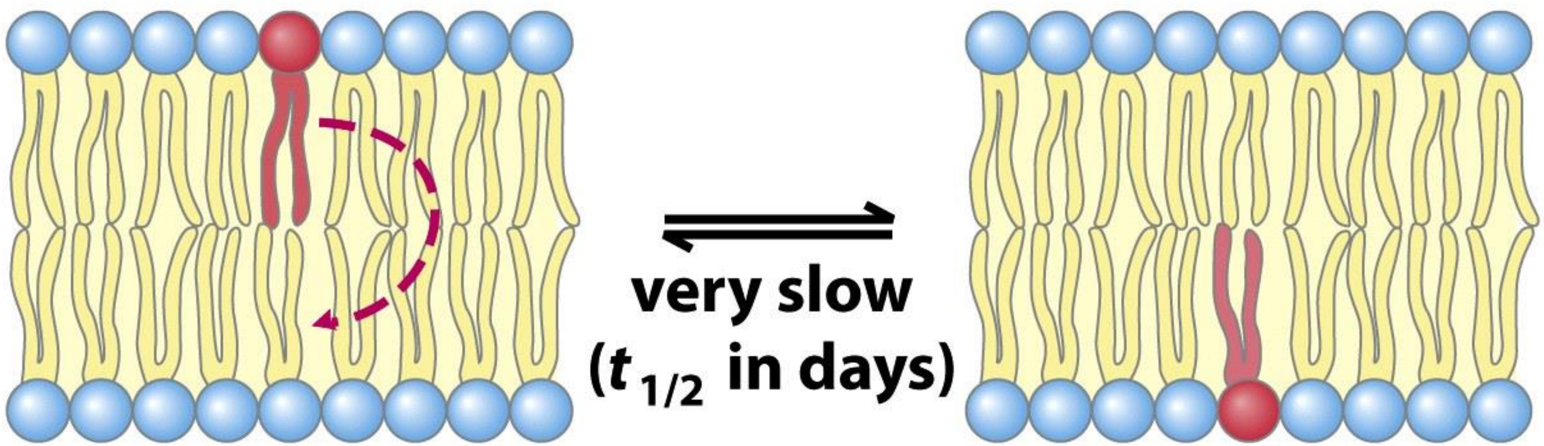
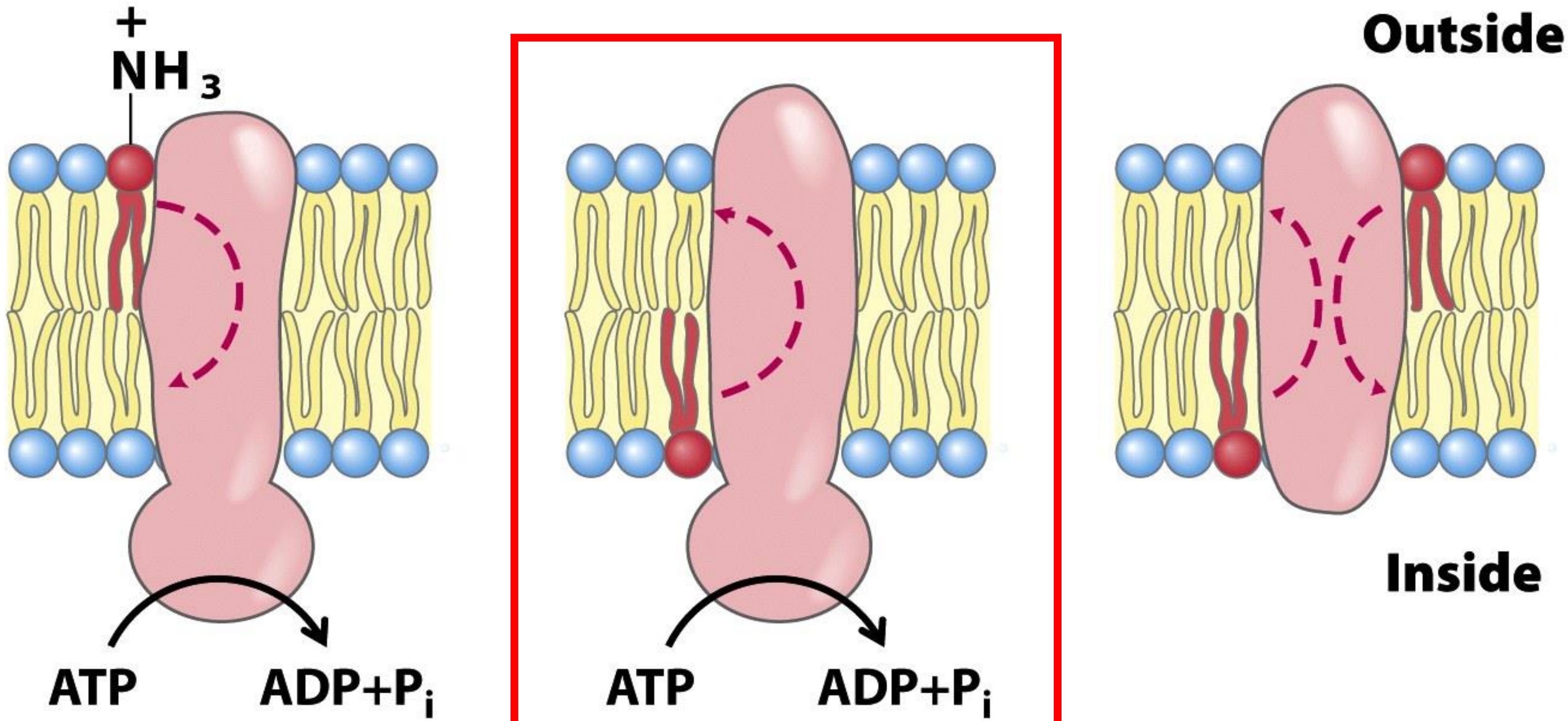


Figure 11-16a
Lehninger Principles of Biochemistry, Fifth Edition

Catalyzed transbilayer translocations



Flippase (P-type ATPase) moves PE and PS from outer to cytosolic leaflet	**Floppase** (ABC transporter) moves phospholipids from cytosolic to outer leaflet	**Scramblase** moves lipids in either direction, toward equilibrium

Figure 11-16c
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Multidrug resistance-associated proteins (MRP1) are floppase. They are responsible for drug resistance of some cancer cells

Biological membranes are composed by:
Polar Lipids and **Proteins**

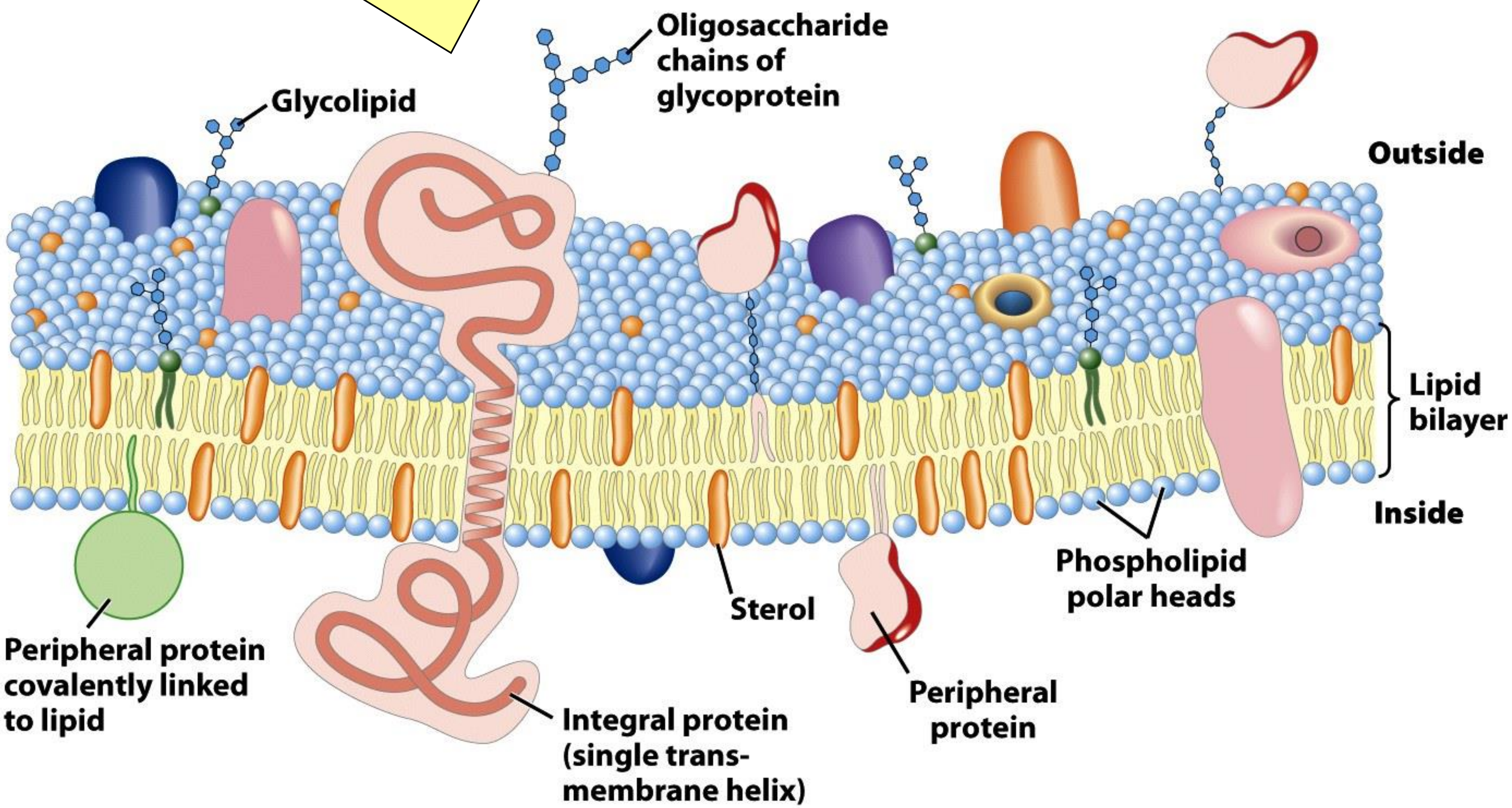

Figure 11-3
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Membrane Proteins: Integral or peripheral

Membrane proteins are responsible for the dynamic processes associated to the plasma membranes, thus specific proteins occur ONLY in particular membranes!

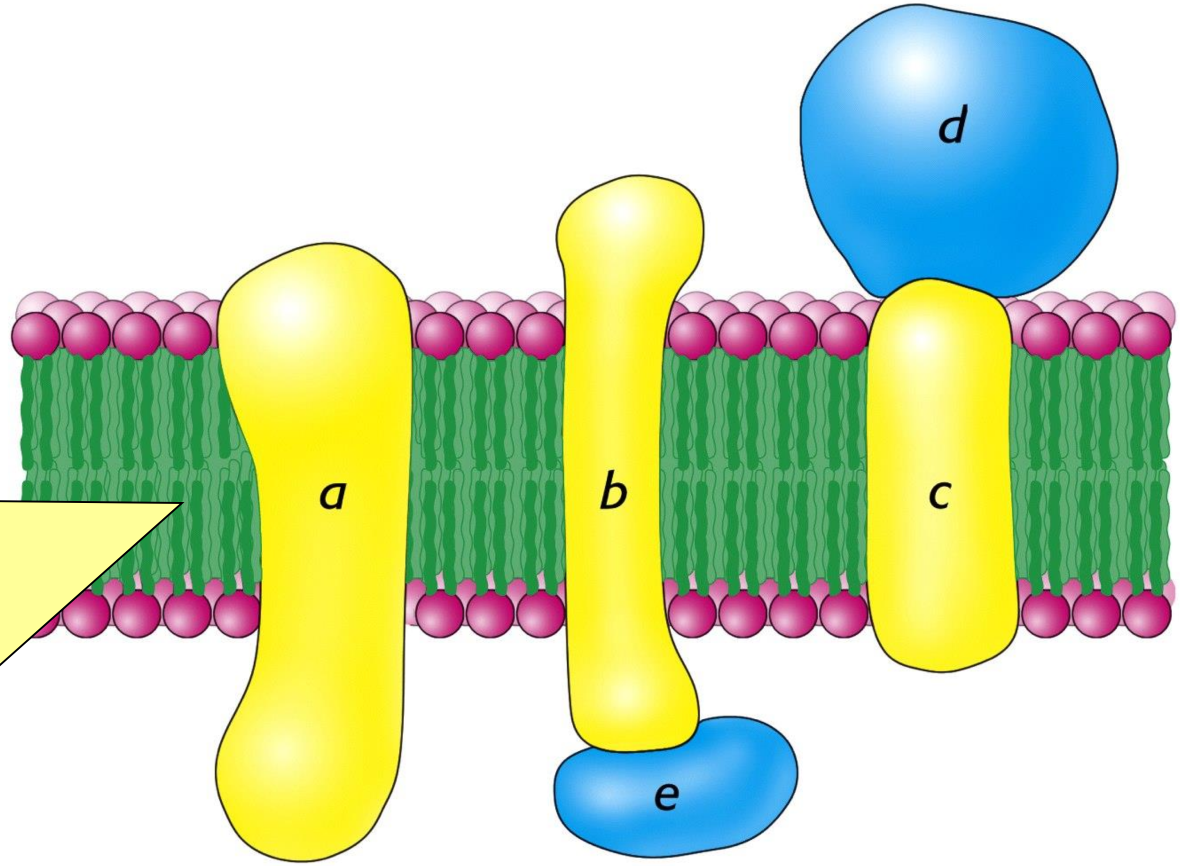

Figure 12-17
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Table 12-4 Compositions of Some Biological Membranes

Membrane	Protein (%)	Lipid (%)	Carbohydrate (%)	Protein to Lipid Ratio
Plasma membranes:				
Mouse liver cells	46	54	2–4	0.85
Human erythrocyte	49	43	8	1.1
Amoeba	52	42	4	1.3
Rat liver nuclear membrane	59	35	2.0	1.6
Mitochondrial outer membrane	52	48	(2–4)[a]	1.1
Mitochondrial inner membrane	76	24	(1–2)[a]	3.2
Myelin	18	79	3	0.23
Gram-positive bacteria	75	25	(10)[a]	3.0
Halobacterium purple membrane	75	25		3.0

[a]Deduced from the analyses.

Source: Guidotti, G., *Annu. Rev. Biochem.* **41**, 732 (1972).

Table 12-4

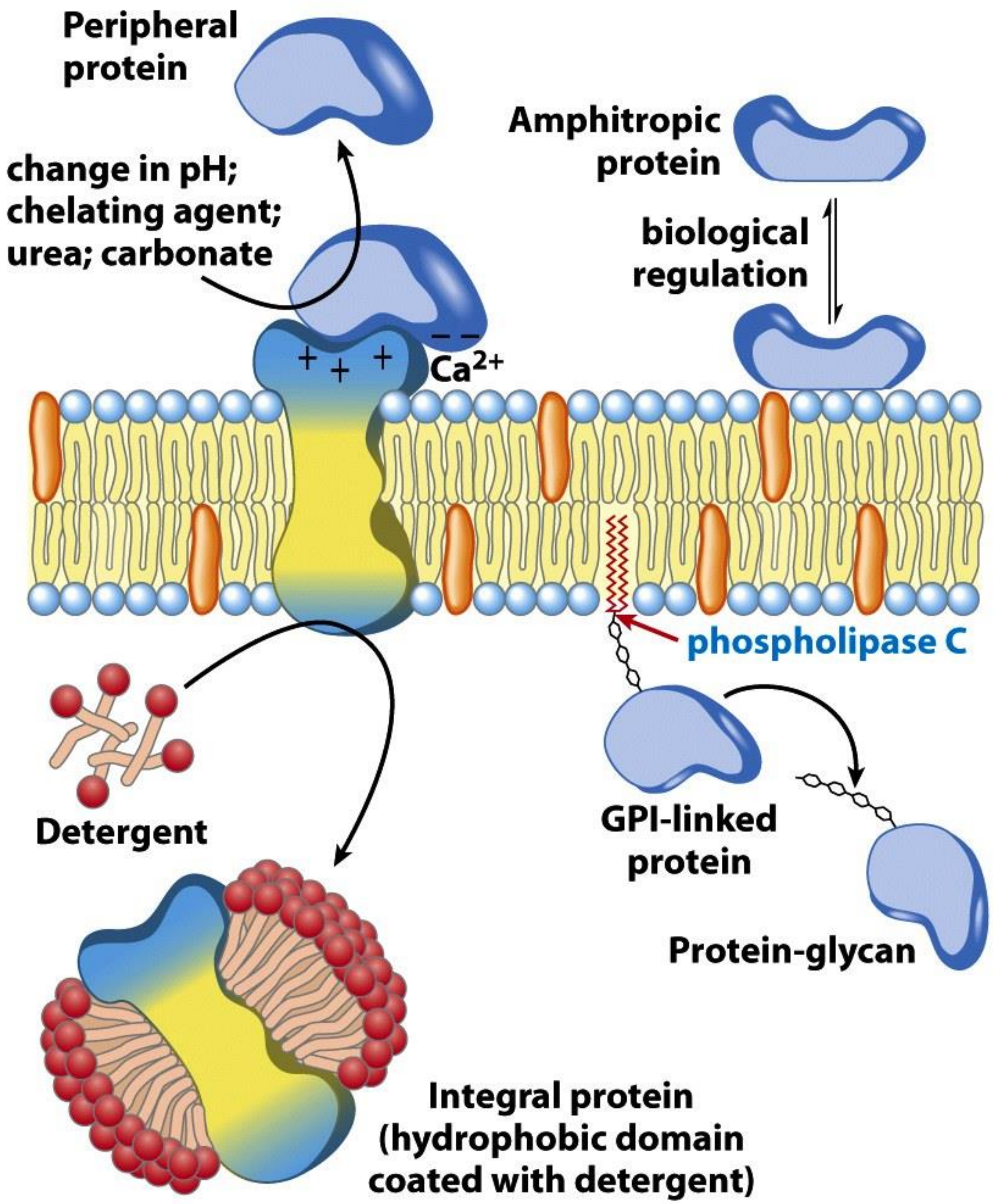

Figure 11-6
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Integral Proteins

The attachment of integral proteins to the membrane is due to hydrophobic interaction between membrane lipids and hydrophobic domains of the proteins.

Membrane spanning α-helices are the most common structural motif in membrane proteins

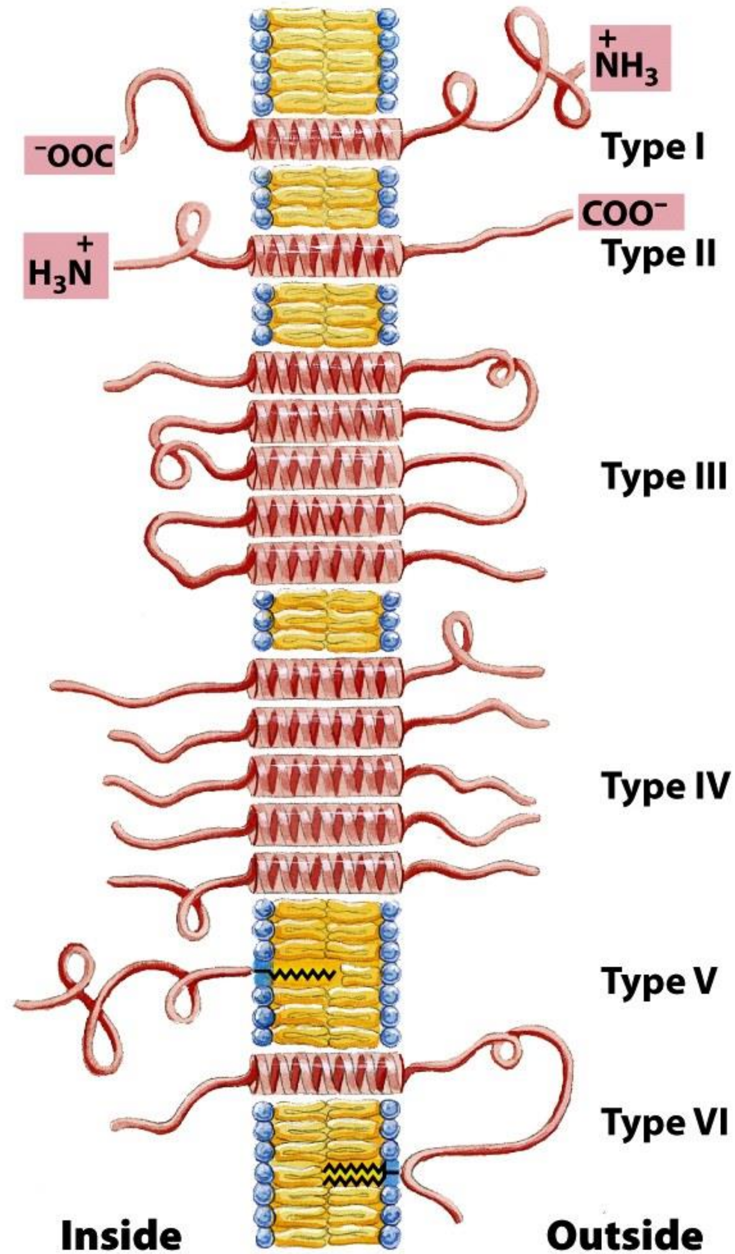

Figure 11-8
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How can we determine if a membrane protein is an integral membrane protein?

From the Sequence (bacteriorhodopsin)

A Q I T G R P E W I W L A L G T A L M G L G T L Y F L V K G M G V S D P D A K K F Y A I T T L V P A
I A F T M Y L S M L L G Y G L T M V P F G G E Q N P I Y W A R Y A D W L F T T P L L L L D L A L L V
D A D Q G T I L A L V G A D G I M I G T G L V G A L T K V Y S Y R F V W W A I S T A A M L Y I L Y V
L F F G F T S K A E S M R P E V A S T F K V L R N V T V V L W S A Y V V V W L I G S E G A G I V P L
N I E T L L F M V L D V S A K V G F G L I L L R S R A I F G E A E A P E P S A D G A A A T S

Figure 12-19
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Hydrophaty Index

Hydrophaty Index measures the free energy changes for the transfer of individual amino acid residues from a hydrophobic to an aqueous environment

Table 12.2 Polarity scale for identifying transmembrane helices

Amino acid residue	Transfer free energy in kJ mol^{-1} (kcal mol^{-1})
Phe	15.5 (3.7)
Met	14.3 (3.4)
Ile	13.0 (3.1)
Leu	11.8 (2.8)
Val	10.9 (2.6)
Cys	8.4 (2.0)
Trp	8.0 (1.9)
Ala	6.7 (1.6)
Thr	5.0 (1.2)
Gly	4.2 (1.0)
Ser	2.5 (0.6)
Pro	−0.8 (−0.2)
Tyr	−2.9 (−0.7)
His	−12.6 (−3.0)
Gln	−17.2 (−4.1)
Asn	−20.2 (−4.8)
Glu	−34.4 (−8.2)
Lys	−37.0 (−8.8)
Asp	−38.6 (−9.2)
Arg	−51.7 (−12.3)

Source: After D. M. Engelman, T. A. Steitz, and A. Goldman. *Annu. Rev. Biophys. Biophys. Chem.* 15(1986):321–353.

Note: The free energies are for the transfer of an amino acid residue in an α helix from the membrane interior (assumed to have a dielectric constant of 2) to water.

Table 12-2
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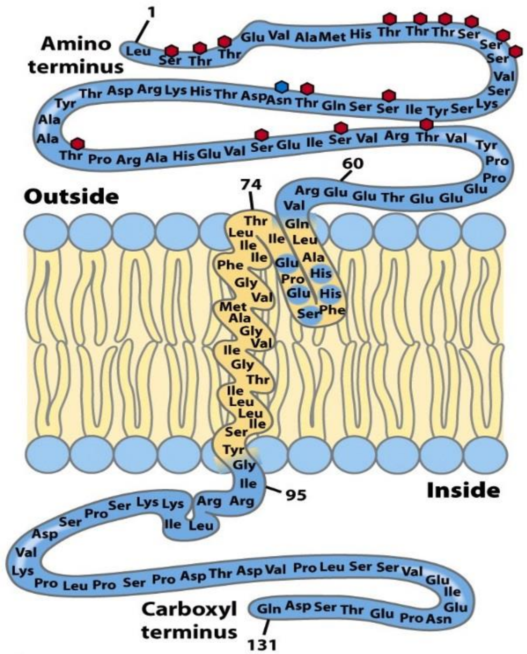

Figure 11-7
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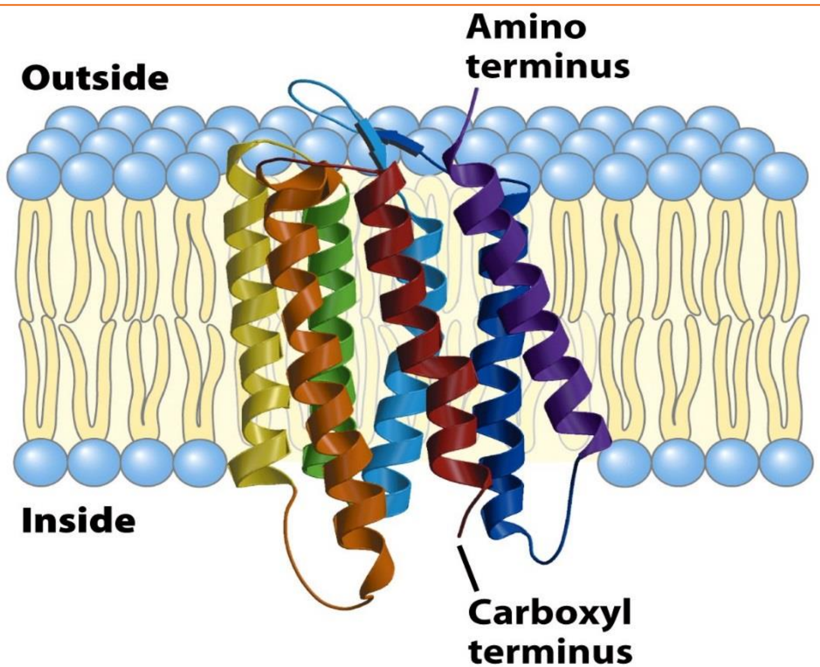

Figure 11-9
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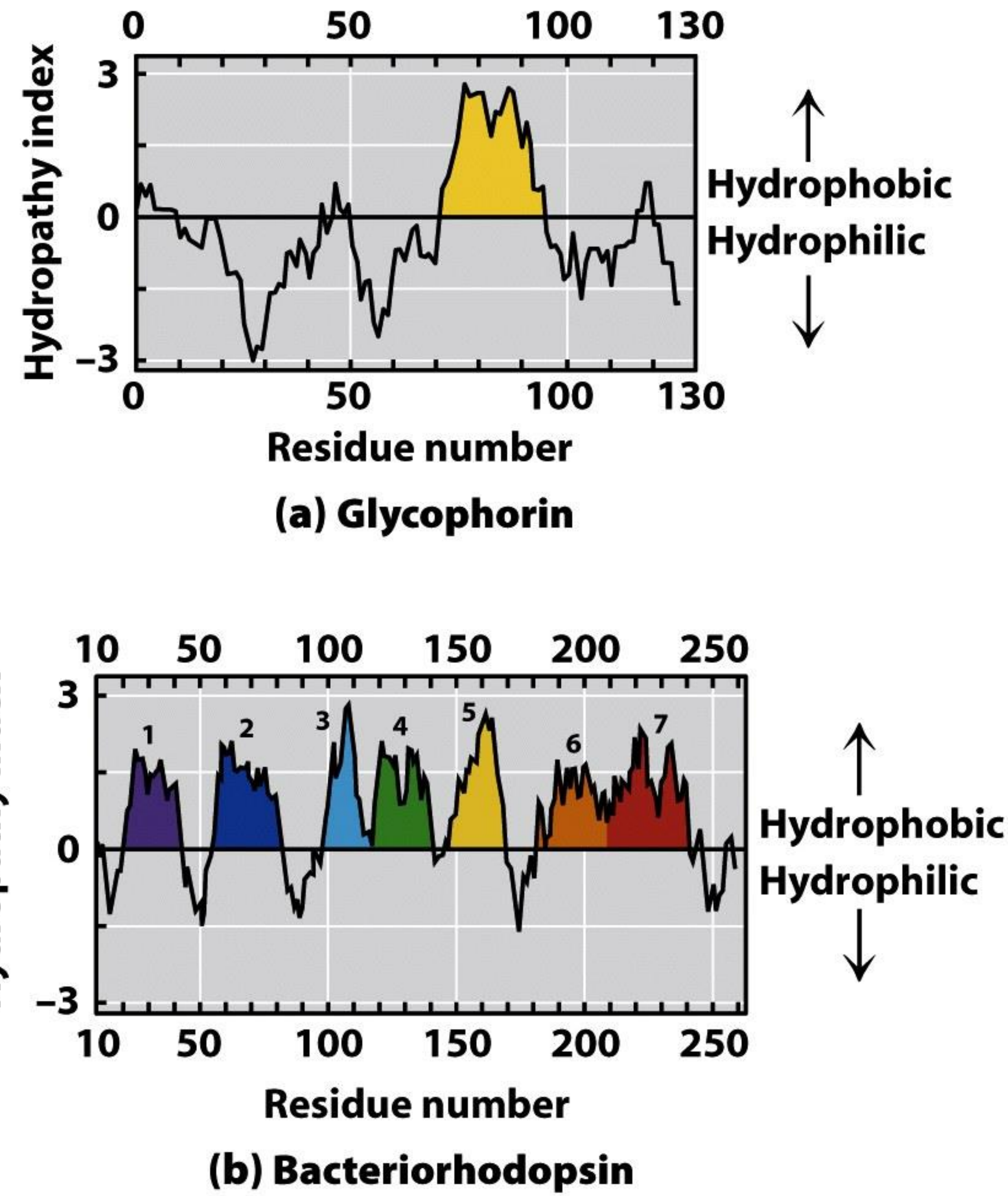

Figure 11-11
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Porin is an example of integral membrane protein with beta barrel structure

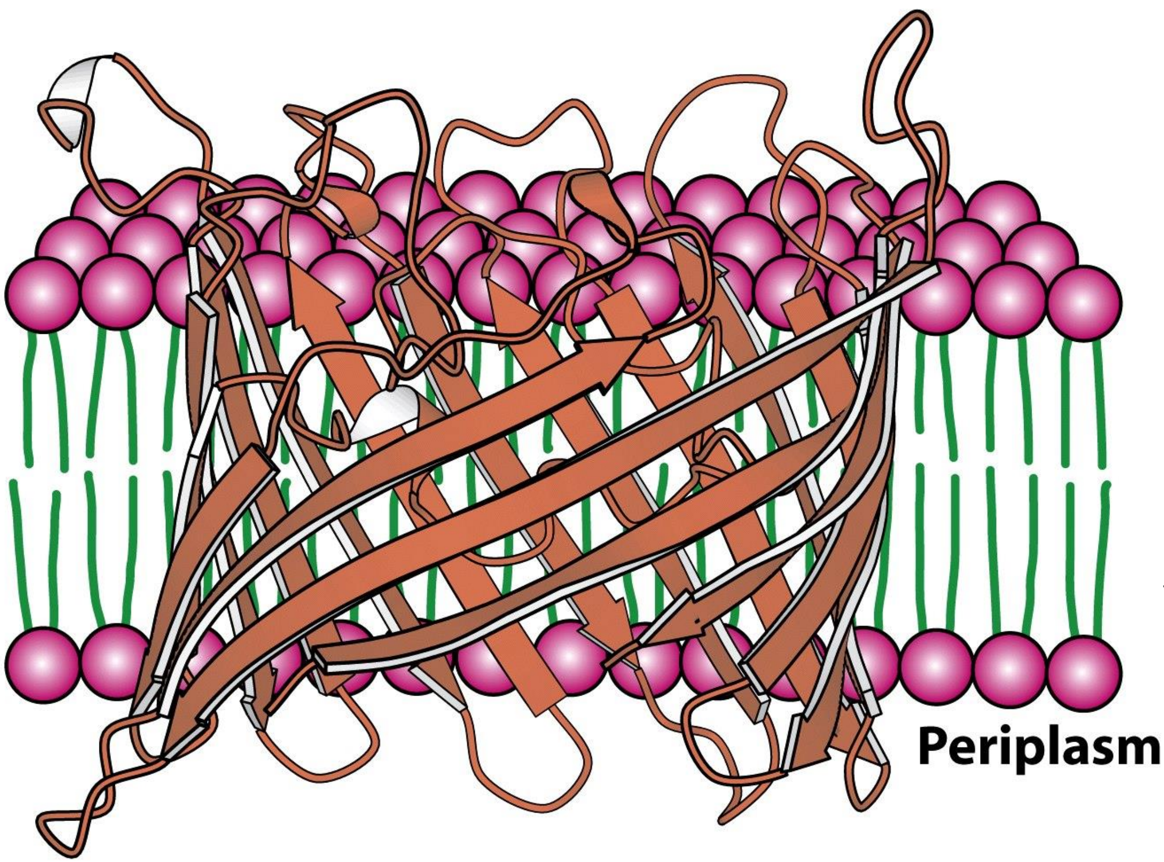

Figure 12-20a
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Integral membrane proteins with β sheet structure generate channels that are non polar outside and polar inside. Their sequence is characterized by the alternation of hydrophobic and hydrophilic aminoacids along the β strand.

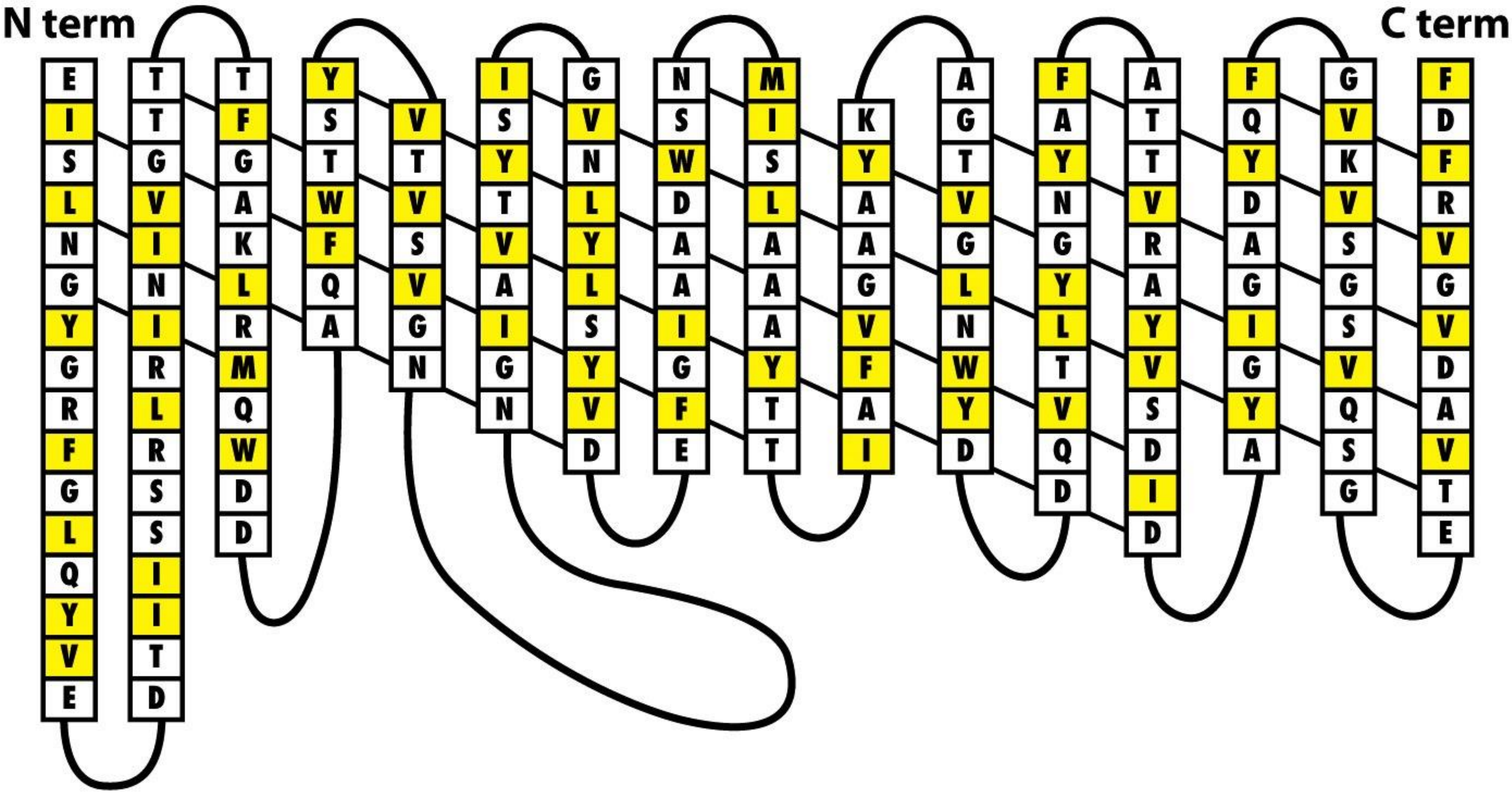

Figure 12-21
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Integral Membrane-binding proteins can interact strongly with the lipid bilayer without crossing it.

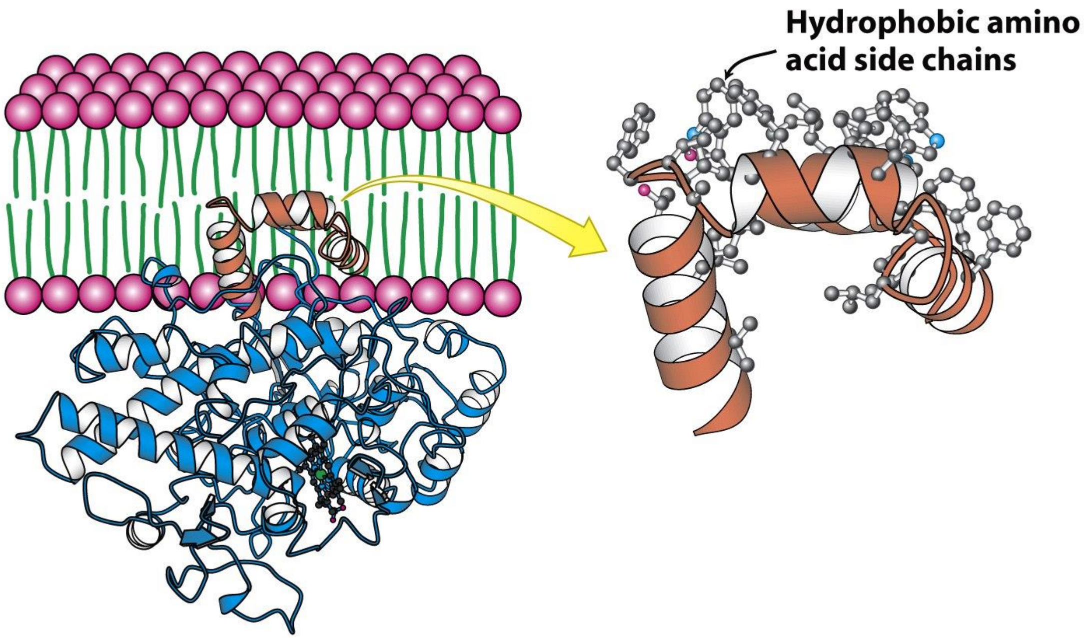

Figure 12-23
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Prostaglandin H_2 synthase-1
Binds the outer surface of the ER membrane

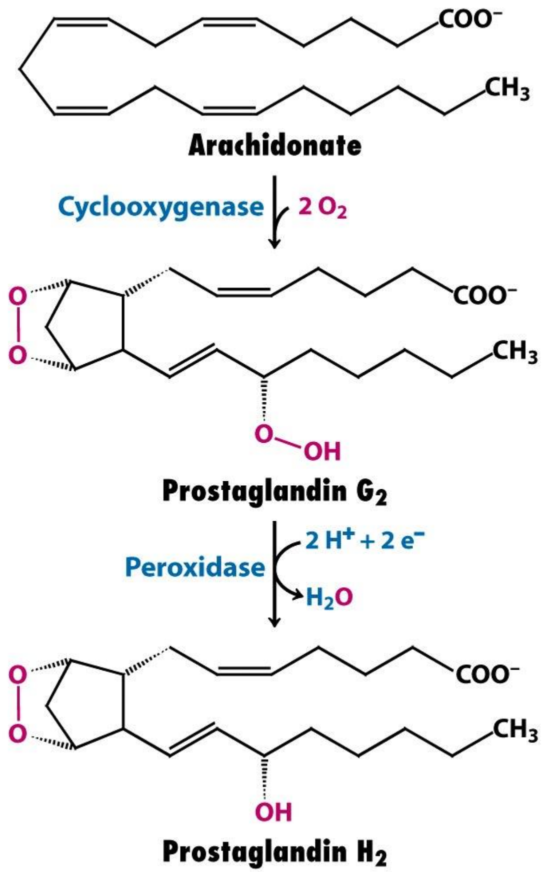

Figure 12-22
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Aspirin action

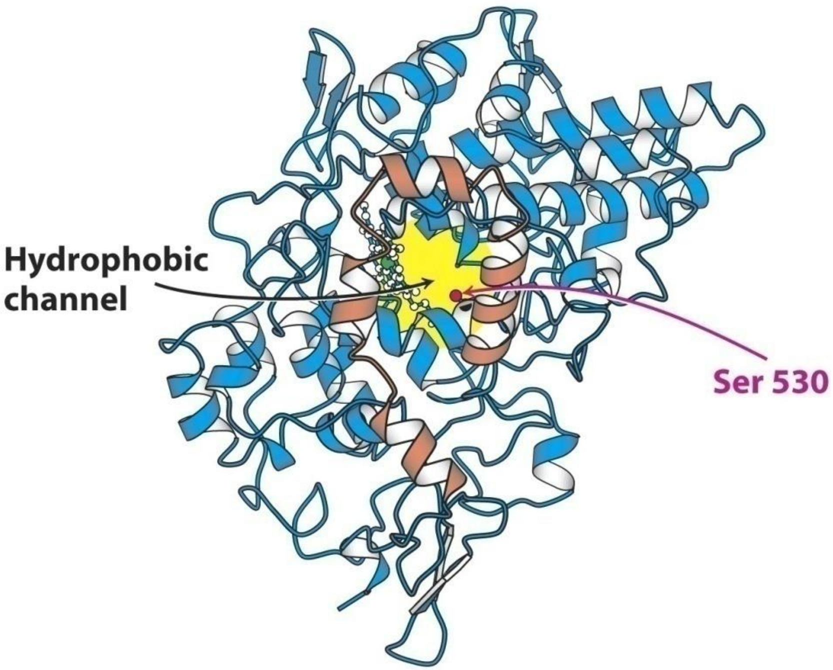

Figure 12-24
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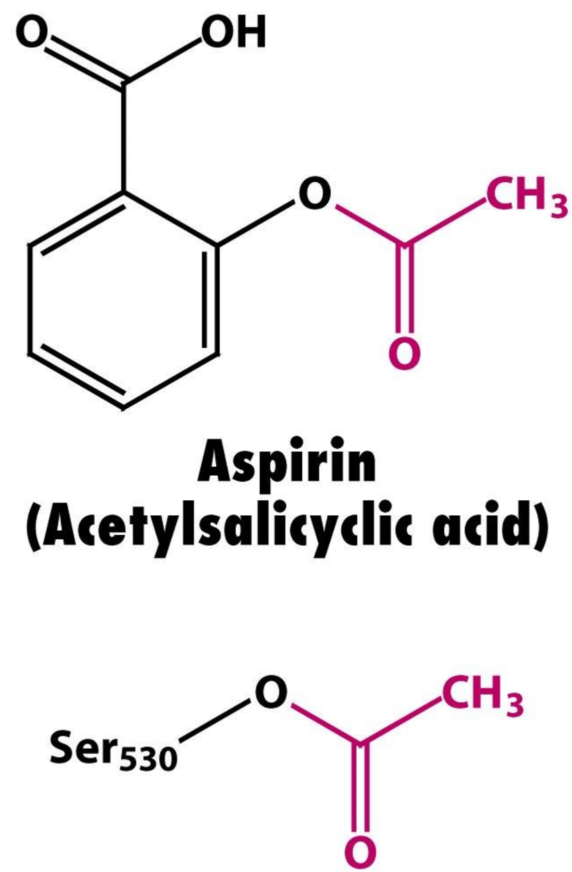

Figure 12-25
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LIPID-LINKED MEMBRANE PROTEINS

Lipid-linked membrane proteins are associated to the membrane in a weaker way with respect to integral proteins. Their association with the membrane is in many cases reversible.

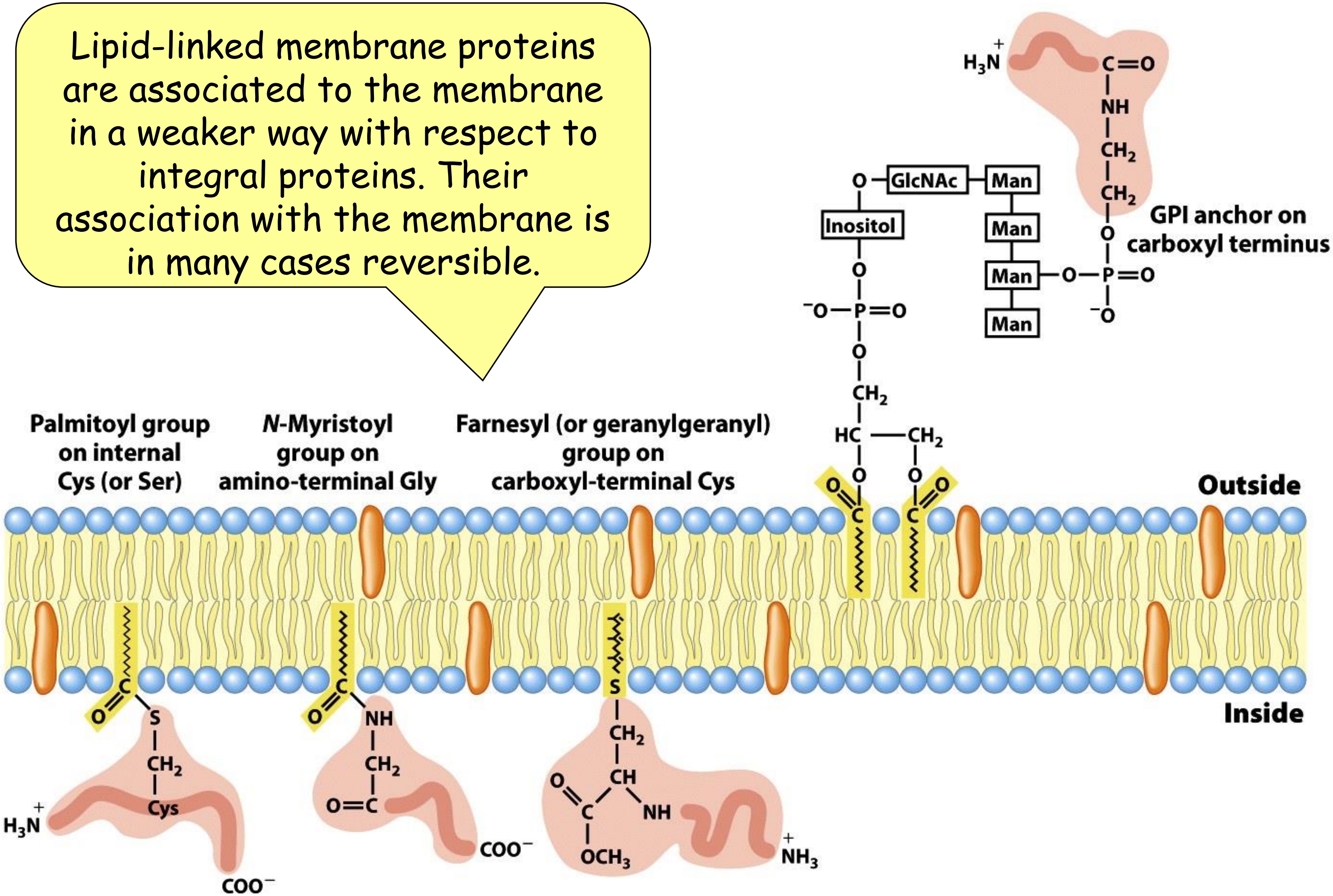

Figure 11-14
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CAN MEMBRANE PROTEINS MOVE IN THE LIPID BILAYER?

DOES THEIR MOVEMENT NEED ENERGY
OR IS SPONTANEOUS?

Some proteins can easly diffuse in the biological membranes as polar lipids do,
But often group of proteins can move only together, meaning they cannot move relative to one another.
i.e. Glycophorin and Chloride-bicarbonate exchange proteins in the erythrocytes membrane.

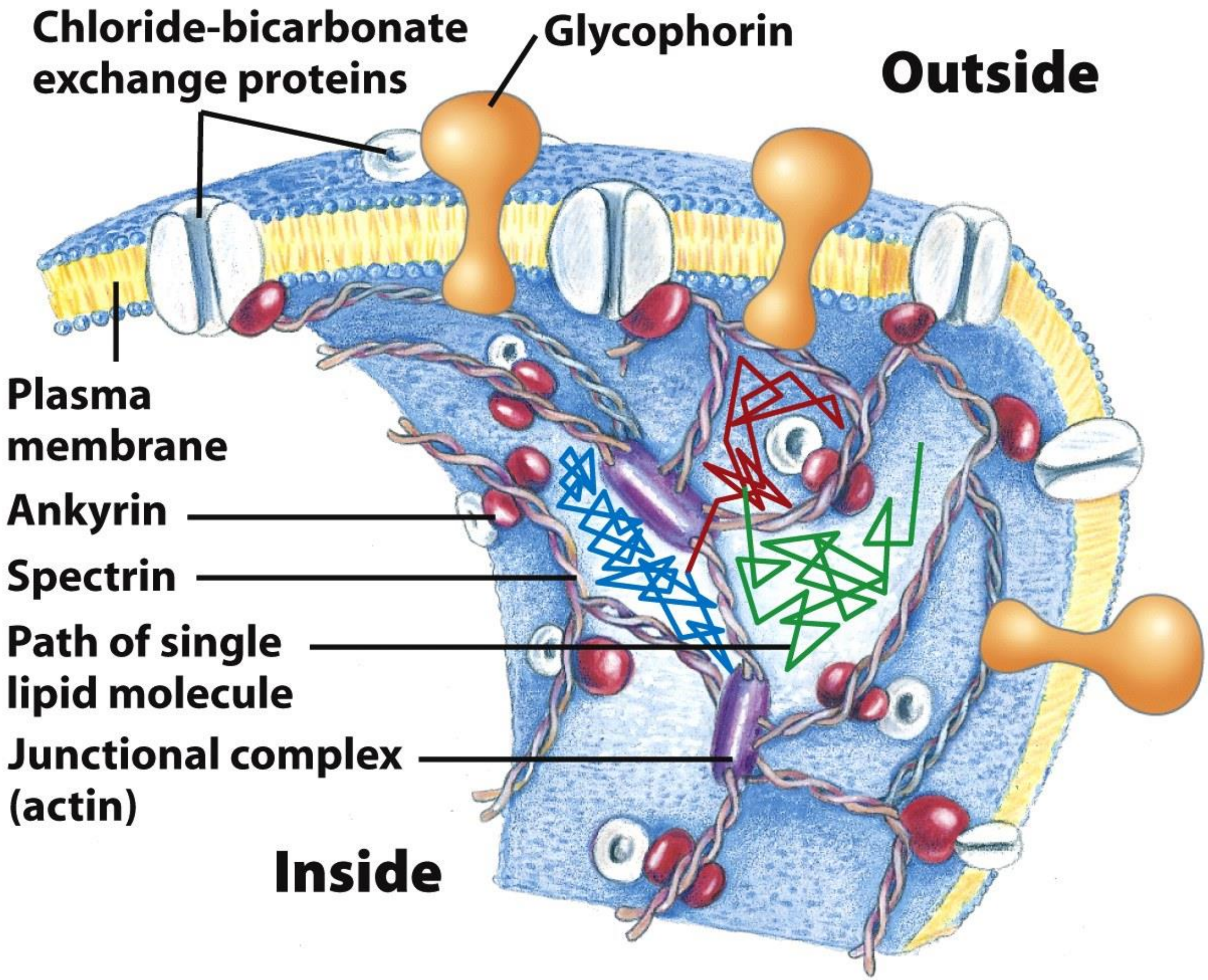

Figure 11-19
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Lipid rafts are more ordered and tightly packed than the surrounding bilayer.
Composition: more saturated acyl chains, presence of cholesterol

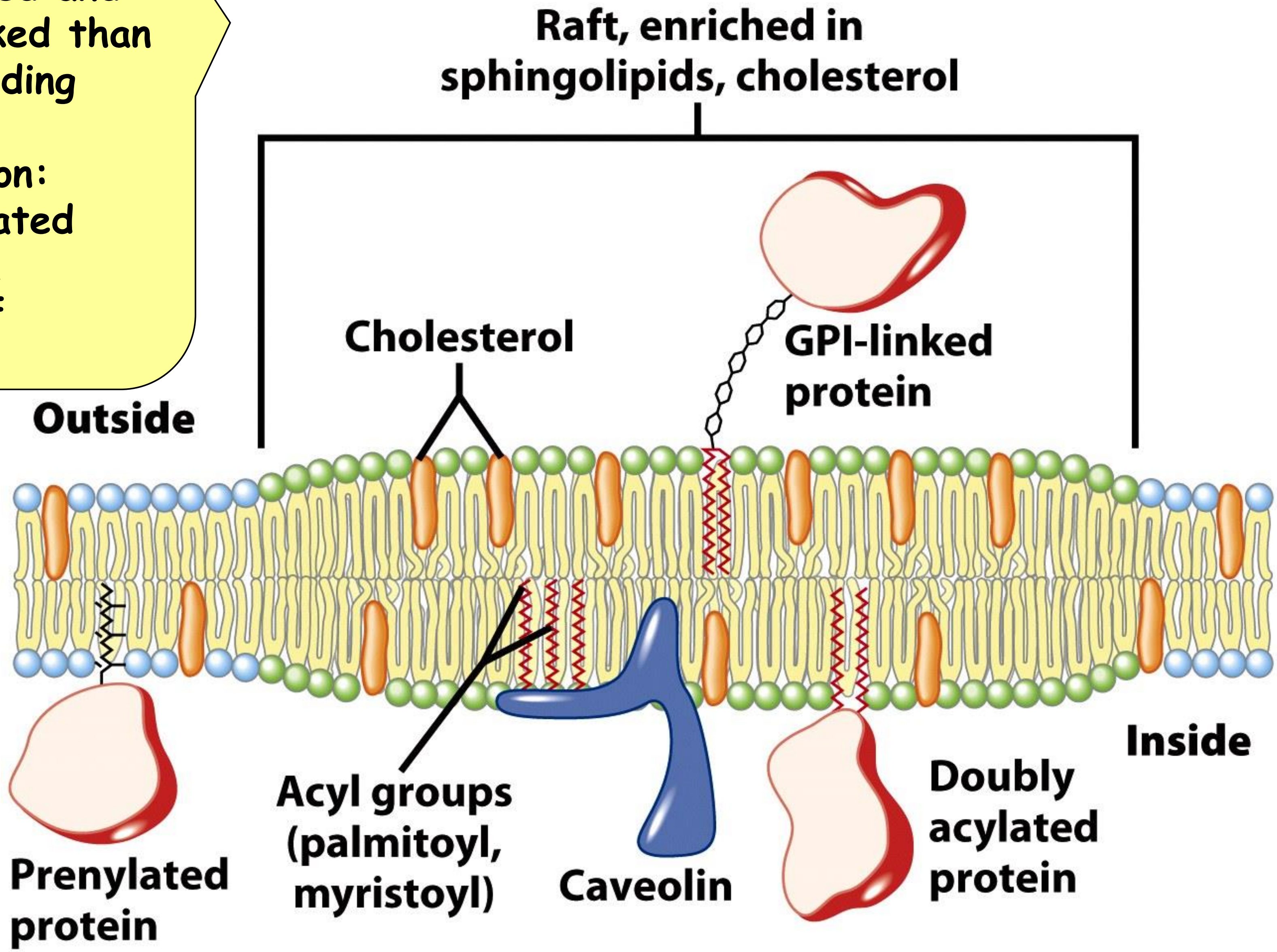

Figure 11-20a
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